THE SPECTRAL SHORTCUT: HOW A FAST CONTEXTUAL PATHWAY SUPPRESSES SPECTRAL LEARNING IN SERIAL SPECTRAL–SPATIAL MODELS

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ABSTRACT

We study how relative module training rates affect spectral adaptation in serial spectral–spatial models, in which a small encoder compresses each pixel's spectrum and a wide spatial network reads the encoded neighbourhood. An exactly solvable serial logistic model bounds the spectral encoder's update at matched fit by a fraction $1/(1 + Mv_0^2)$ of the update a spectral-only model needs, establishes contextual-reversal failure under explicit finite-width conditions, and shows through a $M^{-1/2}$-normalized readout that replicated-readout width acts through the training metric; normalization removes the width dependence but need not prevent the failure. In a nonlinear synthetic model with two cues of matched signal-to-noise ratio, slowing the spatial head by $256\times$ raises the linear accessibility of the spectral cue at matched training loss from 0.66 to 0.85 while the trained classifier remains strongly context-reliant (reversal accuracy 0.12 to 0.14); retraining the head on decorrelated context from the frozen encoders then recovers shifted accuracy of 0.75–0.84 from the slow-head encoders against 0.54–0.62 from the fast-head ones, no better than random encoders. Experiments using measured tissue spectra with constructed context distinguish high and low initial spectral accessibility, show strong contextual reliance in both, and show a weaker rate response at high contextual amplitude. The results separate adaptation of the encoder, accessibility of the spectral cue, reliance of the fitted classifier, and what a specified retraining recovers.

1 INTRODUCTION

Infrared and quantum-cascade-laser histology classify tissue pixel by pixel from a spectrum of several hundred channels, and the models that do so are serial: a small encoder compresses each pixel's spectrum, and a much larger spatial network reads the encoded neighbourhood (Berisha et al., 2019; O'Leary et al., 2026). Practitioners often pretrain or freeze the spectral stage, and a companion tokenization study (anonymous, under review) reports a frozen random encoder competing with a jointly trained one. Such comparisons do not settle whether joint training starves the encoder: the spatial stage can predict the label from the neighbourhood, and a model that learns to do so may never need the spectrum.

This paper asks how the relative training speed of the two modules affects what the encoder learns, and keeps four measurements apart that are easily conflated: adaptation of the encoder, linear accessibility of the spectral cue in its output, reliance of the trained classifier on context, and what a specified retraining procedure can recover afterwards. We prove one instance exactly: in a linear-encoder model with a replicated contextual readout, a faster contextual pathway suppresses the encoder's update at matched fit, the fitted model fails under contextual reversal under explicit finite-width conditions, and readout normalization removes the width dependence (Section 3). We then measure, with interventions on training speed, how each responds in a nonlinear convolutional model on synthetic data (Section 4) and with measured tissue spectra in constructed neighbourhoods (Section 5), each against a control that removes the proposed cause.

Contributions.

- **Theorem 1.** For the serial cross-entropy model with M replicated contextual readouts, gradient flow from general initialization moves the spectral coefficient at the matched fitting margin by at most a fraction $1/(1 + Mv_0^2)$ of the update a spectral-only model needs; the fitted model misclassifies every example under contextual reversal when the initial spectral coefficient is below half the margin and $Mv_0^2 > m/(m-2a_0)$, with expected error tending to $\Phi(m/2\sigma)$ over isotropic Gaussian initialization as $M \to \infty$; and a $M^{-1/2}$-normalized readout removes the width dependence, though it need not remove the failure.
- **Synthetic intervention.** In a ReLU convolutional head over a linear encoder, with two cues of matched signal-to-noise ratio, an informative context suppresses the encoder's spectral learning at matched fit at every width. Slowing the whole head by $256\times$ raises the spectral cue's linear accessibility from 0.66 to 0.85, monotonically in every seed, while the trained classifier's reversal accuracy moves from 0.12 to 0.14; a normalized readout reduces the suppression at large width; an uninformative context removes it. Retraining the head on decorrelated context recovers shifted accuracy of 0.75–0.84 after the slow head and 0.54–0.62 after the fast one, no more than from random encoders.
- **Real spectra.** With measured centre spectra in constructed neighbourhoods, informative-context arms are strongly context-reliant in a high- and a low-accessibility regime; in the latter the encoder's adaptation is limited relative to the uninformative-context control, with a rate response that grows as the cue amplitude falls, and more than 99.9% of the net encoder update lies in the constructed-cue span. Where the spectral cue is accessible, retraining the head recovers 0.90–0.96 shifted accuracy from every encoder, a random one included: the failure was in the use of the spectrum, not its availability. With natural neighbourhoods, on tissue and on a public hyperspectral scene, no competition arises (Appendices E and F).
- **Scope.** Scalar curvature on a production model does not identify the mechanism, and an earlier apparent freezing advantage was protocol-confounded (Appendix G); no mitigation is tested on natural context.

Related work. That one predictive feature can suppress the learning of another is gradient starvation (Pezeshki et al., 2021), and a preference for simpler features that fails under shift is simplicity bias (Shah et al., 2020); modality competition in multimodal training (Wang et al., 2020; Peng et al., 2022; Huang et al., 2022; Du et al., 2023) studies the same imbalance between input streams. That core features can be present in a representation while a spurious feature is used, and that last-layer retraining can then restore robustness, is the observation of Kirichenko et al. (2023); our addition is that the relative training rate of the modules at the original fit changes what such retraining can recover, measured at matched fit against a control. In infrared histology, insensitivity to a learned sixteen-feature spectral bottleneck has been read as showing that tissue classification needs only a small set of spectral features (O'Leary et al., 2026), and CNN classifiers are reported insensitive to the compression method (Müller et al., 2023). Appendix E tests that reading with natural neighbourhoods: a learned twelve-dimensional bottleneck loses nothing, sixteen principal components lose 0.07–0.11 macro-F1 per pixel and 0.04 for the spatial model, and the jointly trained encoder learns the class signal exactly as it does without informative context; compressibility is not redundancy, and a natural 3×3 neighbourhood, being the same class as its centre, is not the competing cue of the theorem. The mathematics uses the balance invariant of gradient flow in layered linear models (Du et al., 2018; Saxe et al., 2014; Yun et al., 2021) next to lazy-training analyses (Chizat et al., 2019; Yang & Hu, 2021); we measure learning outcomes rather than curvature, whose width dependence is normalization-driven (Karakida et al., 2019; van Laarhoven, 2017) (Appendix G); freezing in practice (Zhang et al., 2022; Coil & Cheney, 2025) is the setting, not a remedy we evaluate.

2 Model, task, and what counts as failure

Serial spectral–spatial model. An input is an image whose pixel p carries a spectrum $x_p \in \mathbb{R}^S$ and a label $y_p \in \{-1, +1\}$. A spectral encoder $f_{\boldsymbol{\theta}} : \mathbb{R}^S \to \mathbb{R}^K$ acts on each pixel separately, and a spatial model $g_{\boldsymbol{\phi}}$ of width M reads the encoded neighbourhood of p to produce a logit F_p. The encoder is linear, $f_{\boldsymbol{\theta}}(x) = Wx$ with $W \in \mathbb{R}^{K\times S}$, in the theorem and in the experiments; the spatial model is a replicated readout in the theorem and a ReLU convolutional head in the experiments. Training minimizes the mean margin loss $\mathcal{L} = \frac{1}{N}\sum_p \log(1 + e^{-y_p F_p})$ by gradient flow or plain

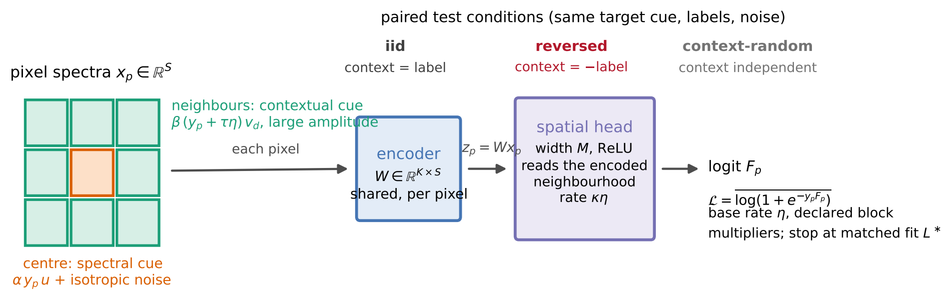

Figure 1: The serial model and the two cues. Each pixel's spectrum passes through the shared encoder; the spatial model reads the encoded neighbourhood (a 5×5 receptive field from two 3×3 convolutions in Experiment 2; one nine-spectrum patch in Experiment 3). The contextual cue lies in the eight nearest neighbours along large-amplitude directions that a random encoder passes; the spectral cue lies in the pixel's own spectrum along a small-amplitude direction that, in the low-accessibility settings, requires encoder alignment. Evaluation under iid, reversed and context-random contexts measures reliance and failure.

gradient descent with a base rate η; the baseline uses that one rate for every parameter, and the interventions multiply the rate of a named block by a stated factor, so that every difference in training speed between the modules is either a property of the model or an explicit, declared multiplier.

Two cues. The training distribution carries the label in two places. The *spectral cue* lies in the pixel's own spectrum along a direction the encoder must discover: a randomly initialized encoder attenuates it. The *contextual cue* lies in the neighbourhood along directions of large amplitude: a randomly initialized encoder passes it, so a trained readout can use it at once. The two cues are calibrated to matched discriminant accuracy on calibration data (a matched signal-to-noise construction in the synthetic case; a training-side estimate on real spectra), so that a preference for one cue is not simply a preference for the more predictive one. The theorem allows any (a_0, v_0); the experiments select its regime of interest, a small initial spectral contribution and an initially accessible contextual one, and measure that accessibility asymmetry directly (Section 4).

Matched fit, spectral learning, reliance, failure. We compare models at their first attainment of a prespecified training-fit threshold: a common margin in the solvable model and mean training loss in the experiments. Spectral learning means adaptation of the designated spectral coefficient in the theorem and increased independently measured linear accessibility of the centre's spectral cue in the experiments. We measure contextual reliance by changing the assigned contextual signal while preserving the target's spectral cue, label and noise, and recording the resulting change in predictions or risk. We call that reliance a failure when the shifted risk exceeds that of a comparator of the same architecture trained with an uninformative (decorrelated) context to the same fitting threshold. The theorem supplies an exact reversal failure against a spectral-only model; the noisy experiments measure the corresponding risk difference against that trained comparator, without assuming a zero-error reference. Four measurements are kept distinct throughout: adaptation of the encoder, linear accessibility of the spectral cue on clean spectral inputs (the probe), use of context by the original predictor, and performance after an explicit retraining procedure.

Test conditions. Every evaluation family is generated from the same labels and noise as *iid* (context agrees with the label), *reversed* (the assigned context carries the opposite label) and *context-random* (an independent label: context present but uninformative). In the synthetic experiment the intervention changes the contextual field at every pixel while the target's spectral component, label and noise are shared; in the real-spectrum experiment each centre spectrum is held fixed. Reversal measures reliance; context-random measures what the predictor does from the spectrum in the presence of context.

3 MAIN RESULT: SELECTIVE SPECTRAL LEARNING AT MATCHED FIT

We state the mechanism in the simplest serial model that has both cues, a shared encoder, and a replicated contextual readout. Training inputs are two sites: the pixel's own spectral coordinate S and a contextual coordinate C, both equal to the label. The encoder $W = (a, v)$ reads each site with one coefficient; the head adds the spectral coordinate directly and the contextual one through M replicated readouts. Replication is the only way width enters. The statement combines the serial instance, its general-initialization corollary and the readout-normalization proposition of Appendix A.

Theorem 1 (Selective spectral learning at matched fit)**.** *Let Y be uniform on $\{-1, 1\}$, $x_1 = (S, 0)$, $x_2 = (0, C)$, and $S = C = Y$ in training. The shared encoder $W = (a, v)$ produces $z_1 = aS, z_2 = vC$; the head is $F = z_1 + bz_2$, $b = \sum_{j=1}^{M} \beta_j$. For integer $M \geq 1$, train (W, β) by unit Euclidean gradient flow on $\mathcal{L} = \mathbb{E}\log(1 + e^{-YF})$, without other optimizer terms, from fixed $a_0 \in \mathbb{R}$, $v_0 \neq 0$, $\sum_j \beta_{j0} = 0$. Fix $m > 0$ and $T_m = \inf\{t \geq 0 : q_J(t) \geq m\}$, $q_J = a + bv$; thus $T_m = 0$ if $a_0 \geq m$.*

(a) Suppression. *If $a_0 < m$, set $\Delta = m - a_0$, $u_M = a(T_m) - a_0$. Then $T_m < \infty$ and*

$$0 < \frac{u_M}{\Delta} \leq \frac{1}{1 + Mv_0^2}, \qquad Mu_M \longrightarrow \frac{\Delta}{v_0^2} \quad (M \to \infty).$$

The spectral-only head $F = aS$ requires update Δ. If q_F instead trains only β from the same initialization with W frozen, then $q_J \geq q_F$ on $[0, T_m]$ and $\sup_{0 \leq t \leq T_m}(q_J - q_F) \leq C_0/(p_0 M v_0^2)$, where $C_0 = 1 + 2\Delta^2/v_0^2$ and $p_0 = (1 + e^{-a_0})^{-1}$.

(b) Reversal. *On $S = Y, C = -Y$, the stopped margin is $2a(T_m) - m$ when $a_0 < m$, and a_0 otherwise. Error is one if $a_0 < m/2$ and $Mv_0^2 > m/(m - 2a_0)$; it is zero for $a_0 \geq m/2$. The spectral-only comparator has zero error for every a_0. For $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I_2)$, $\sigma > 0$, with zero-sum readouts and the same stopping rule, $\lim_{M\to\infty} \mathbb{E}_{W_0} \mathrm{Err}_{\mathrm{rev}} = \Phi(m/(2\sigma))$, where Φ is the standard normal CDF.*

(c) Normalization. *Replace b by $M^{-1/2}\sum_j \gamma_j$, with $\sum_j \gamma_{j0} = 0$ and the same (a_0, v_0) and unit rates. The aggregate (a, v, b) dynamics equal those at $M = 1$ for every width.*

Remark 1 (Special initialization). For $(a_0, v_0) = (0, 1)$, the scalar-logit GGN blocks $G_{\boldsymbol{\theta\theta}}, G_{\boldsymbol{\phi\phi}}$, $\boldsymbol{\theta} = (a, v)$, $\boldsymbol{\phi} = \beta$, give $\mathcal{D}_{\mathrm{curv}}(0) = M$ (generally Mv_0^2), and $\Psi(m)/(M + 1 + 2m^2) \leq T_m \leq \Psi(m)/(M + 1)$, where $\Psi(m) = m + e^m - 1$.

Proof sketch. Dividing the flow by $\dot{a} = \varrho > 0$, $\varrho = (1 + e^{q_J})^{-1}$, gives $dv/da = b$ and $db/da = Mv$, hence with $u = a - a_0$

$$v = v_0 \cosh(\sqrt{M}u), \qquad b = \sqrt{M}v_0 \sinh(\sqrt{M}u), \qquad q_J = a_0 + u + \tfrac{\sqrt{M}v_0^2}{2}\sinh(2\sqrt{M}u),$$

the balance invariant of gradient flow (Du et al., 2018). The fitting equation $q_J(u_M) = m$ is strictly increasing in u and $\sinh x \geq x$ gives the update bound and its limit; a transformed-margin comparison with the frozen path gives the finite-horizon gap. The sign of the reversed margin $2a(T_m) - m$ and dominated convergence give (b). For (c), differentiating the aggregate readout gives $\dot{b} = v\varrho$, independent of M. Full proofs are in Appendix A.

What the theorem says, and what it does not. Replication acts as an effective learning rate on the contextual readout. At the matched margin the spectral coefficient has moved by at most a fraction $1/(1 + Mv_0^2)$ of the update the spectral-only model needs; the coefficient itself can be informative at initialization and remain so. It is the *adaptation* that is suppressed. When the initial spectral coefficient is below half the margin and the width is large enough, $Mv_0^2 > m/(m - 2a_0)$, the fitted joint model relies on context strongly enough that contextual reversal misclassifies every example, while a spectral-only comparator trained to its own hitting time of the same margin is correct; at small Mv_0^2 the fitted model can be correct even with $a_0 < m/2$, so the failure is a finite-width statement, not a universal one. The spectral-only comparator is trained to its own hitting time; the frozen comparator is evaluated at the joint path's elapsed times through T_m. Part (c) removes the width dependence only: the normalized system is the $M = 1$ system, which for $(a_0, v_0) = (0, 1)$ still fails under reversal for every $m > 0$. Finally, training each contextual readout at rate κ instead of unit rate gives $\dot{b} = \kappa M v\varrho$ and the bound $1/(1 + \kappa M v_0^2)$: the relative training speed of the contextual pathway, not width as such, is the competition parameter the experiments manipulate.

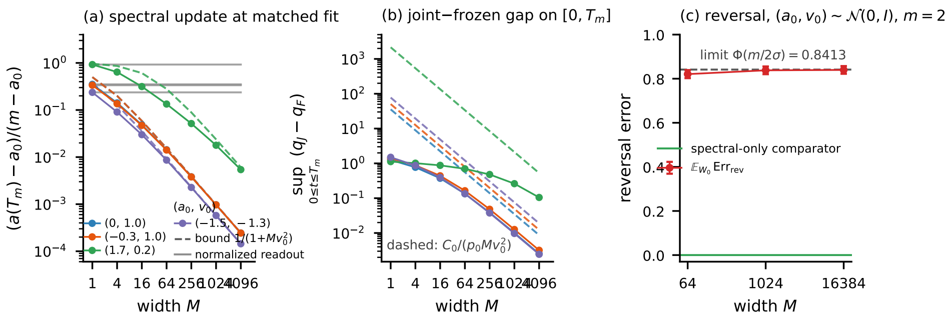

Figure 2: The exact mechanism (Theorem 1; $m = 2.9$ unless stated). (a) Spectral update fraction u_M/Δ at the matched margin against width M for four initializations (a_0, v_0) (solid) with the bound $1/(1 + Mv_0^2)$ (dashed) and the normalized readout of part (c) (grey, flat at the $M = 1$ value). (b) Joint-minus-frozen margin gap $\sup_{0\le t\le T_m}(q_J - q_F)$ with the bound $C_0/(p_0 M v_0^2)$ (dashed). (c) Expected reversal error over $(a_0, v_0) \sim \mathcal{N}(0, I_2)$ against M for $m = 2$, with the limit $\Phi(m/2)$; the spectral-only comparator has error zero.

Numerical check. The closed forms agree with a full $(2 + M)$-parameter integration over 525 initialization–width–margin cases to a relative discrepancy of 1.3×10^{-11} in the endpoint quantities, with no bound violated, and the Gaussian sweep approaches its limit $\Phi(1)$ (0.8207, 0.8380, 0.8397 at $M = 64$, 1024, 16384; Appendix B). This checks the algebra, not the claim beyond the model.

4 Exact-model check and synthetic intervention

The exact model is verified numerically in Section 3; the question here is whether the mechanism survives when the cues are noisy, the routing is learned and the head is nonlinear.

Experiment 2: design. Pixels carry i.i.d. labels $y_p \in \{-1, +1\}$ on 16×16 images with $S = 256$ spectral channels, so the spectral cue cannot be denoised by averaging neighbours. The spectral cue is $\alpha y_p u$ in the pixel's own spectrum, buried in isotropic noise of unit variance, with u a fixed unit direction the encoder must discover. The contextual cue places y_p in each of the eight neighbours of p, along eight fixed orthogonal directions $v_1, \ldots, v_8 \perp u$ of amplitude $\beta = 5$ with context noise $\tau\eta$: the information about y_p sits only in the neighbourhood, because p's own v-coordinates carry its neighbours' labels. Matched signal-to-noise: $\alpha = 1.645$ gives an oracle spectral accuracy of 0.95, and $\tau = 1.70$ is calibrated so that the oracle contextual accuracy (a linear discriminant on the 3×3 window of the eight projections) is 0.950; both values are recorded before any training run (Appendix C). Readiness: through a random encoder $W \in \mathbb{R}^{12\times256}$ the same discriminant reads the contextual cue at 0.905–0.914 and the spectral cue at 0.608–0.653 (three seeds), because the random projection preserves the signal-to-noise ratio of a cue whose noise shares its direction and attenuates by about $\sqrt{K/S}$ one whose noise is isotropic. This is the nonlinear analogue of (a_0 small, $v_0 = O(1)$): the contextual cue is initially accessible to a trained readout; the spectral cue is substantially less accessible. The model is the linear encoder followed by a ReLU convolutional head (3×3 convolution to width M, ReLU, 3×3 convolution to one logit, circular padding; a 5×5 receptive field), trained by full-batch gradient descent on 256 images (65,536 pixels) with base rate $\eta = 10^{-3}$, chosen as the largest rate with a monotone loss at the widest head; the baseline applies it to every parameter, the rate interventions multiply a named block's rate. Runs are compared at the first step with training loss below $L^* = 0.30$, with further snapshots at $0.6, 0.5, 0.4, 0.2, 0.15, 0.10$.

Arms and measures. Standard parameterization at widths $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; a normalized readout ($M^{-1/2}\gamma$ with $\gamma = O(1)$ and the same function at initialization, the control of Theorem 1(c), which under plain gradient descent equals a readout trained at rate $1/M$); training with an uninformative context (the same cue carrying an independent label: the proposed cause removed) at $M \in \{2, 8, 128, 512\}$; a readout learning-rate multiplier and, following the design review, a whole-head multiplier $\kappa \in \{1, 1/16, 1/256\}$ at $M = 32$, which in the theorem gives the bound $1/(1 + \kappa M v_0^2)$; and a frozen encoder. Three seeds each. At every snapshot we record the probe (a discriminant on the encoder output of clean spectral-only inputs, fitted and evaluated on disjoint independent sets; initial values 0.647, 0.658, 0.615, reported as the gain over the run's own initial value), its exact population counterpart $h_u(W) = \|P_{\mathrm{row}(W)} u\|^2$ (optimal probe accuracy $\Phi(\alpha\sqrt{h_u})$; 0.055, 0.060, 0.029 at initialization), the encoder's displacement, and accuracies on paired test sets sharing labels and noise (iid, reversed and context-random change only the contextual field; spectral-only removes it; context-only removes the spectral signal). A recovery procedure was fixed before any shifted result was inspected: the encoder is frozen at its L^* checkpoint, a fresh head with a paired initialization is trained on newly sampled context-random images for 20,000 steps, and the final head is evaluated on the same paired test family.

Results: suppression. In the standard-parameterization baseline every model with an informative context leaves the encoder's spectral accessibility near its initial value at L^*: the probe gain is 0.04 at $M = 2$ and 0.002 at $M = 2048$ (means over three seeds; h_u $0.09 \to 0.05$ against a mean 0.048 at initialization), accuracy under reversal is 0.12–0.13, and accuracy with an uninformative context is at chance (0.50). The same model trained with an uninformative context aligns its encoder before reaching the same loss (probe gain 0.25–0.30; h_u 0.56–0.86) and survives reversal (0.88–0.90). A frozen encoder gives reversal accuracy 0.12–0.14 (Table 4).

Results: the registered predictions. The original width-based prediction was not met at $L^* = 0.30$: on the original five widths only one seed shows the required rank correlations for both alignment and reversal accuracy. On those widths, the normalized readout's alignment and probe flatness met the stated two-of-three-seed criterion, but its reversal flatness did not; its high-width alignment improvement over the standard readout held in every seed. On the amended control grid, the uninformative-context arm failed both the energy-fraction cutoff and the flatness requirement, although it produced substantially higher probe accuracy than informative-context training at every width. The readout-rate sweep supported the alignment prediction and changed reversal accuracy in the opposite direction; frozen encoders remained far below chance under reversal; the whole-head-rate prediction added at the design review held for probe accessibility and row-space retention in every seed, while its reversal rank criterion failed in one seed. Low primary-threshold gains with the stronger labelled secondary trend at loss 0.15 are consistent with contextual fitting preceding spectral adaptation, and neither falsify nor confirm competition (criterion table in Appendix C).

Results: where the width and speed dependence appears. At the labelled secondary threshold 0.15, spectral learning decreases with width in every seed (probe gain 0.22 at $M = 2$, 0.12 at $M = 8$, 0.06 at $M = 128$, 0.03 at $M = 2048$; Spearman -0.86, -0.96, -0.86), and the normalized readout is less suppressed at large width (0.10 at $M = 2048$ against 0.03; Table 5). The whole-head multiplier moves the same quantity at L^* itself: reducing the head's rate by 256 raises mean probe accuracy from 0.662 to 0.851 (paired gains 0.195, 0.178, 0.192) and h_u from 0.067 to 0.406, monotone in every seed (Table 7), in the direction of the theorem's bound $1/(1 + \kappa M v_0^2)$, while remaining below the uninformative-context control (0.89–0.94). Two further runs at $\kappa \le 1/4096$ on one seed did not reach L^* within the step budget (final losses 0.32 and 0.35, $h_u = 0.69$–0.70); they are reported as budget endpoints, not matched-fit points. The readout-only multiplier acts in the same direction but weakly (Table 8).

Results: the original predictor stays context-reliant. Across every matched informative-context run, accuracy under reversal at L^* lies between 0.10 and 0.15 and accuracy with an uninformative context at chance, for every width, the normalized readout and both rate multipliers; the rate change that raises the probe by 0.19 raises reversal accuracy by 0.02 ($0.121 \to 0.143$). Among the tested interventions only training without an informative context removes the reliance: the classifier's use of context and the encoder's accessibility respond very differently to the same intervention.

Results: what retraining recovers. The recovery procedure makes the accessibility difference operational. From the $\kappa = 1$ encoders the retrained head reaches accuracy 0.62, 0.59, 0.54 under reversal (0.61, 0.59, 0.55 with context-random, 0.59 iid), no better than the same procedure on the random initial encoders (0.60, 0.59, 0.53); from the $\kappa = 1/256$ encoders it reaches 0.84, 0.77, 0.75 (0.84, 0.77, 0.76; 0.79 iid), while the original classifiers of the two arms had the same poor shifted performance (0.12–0.15; Table 9). Within this protocol the contextual pathway's relative rate at the

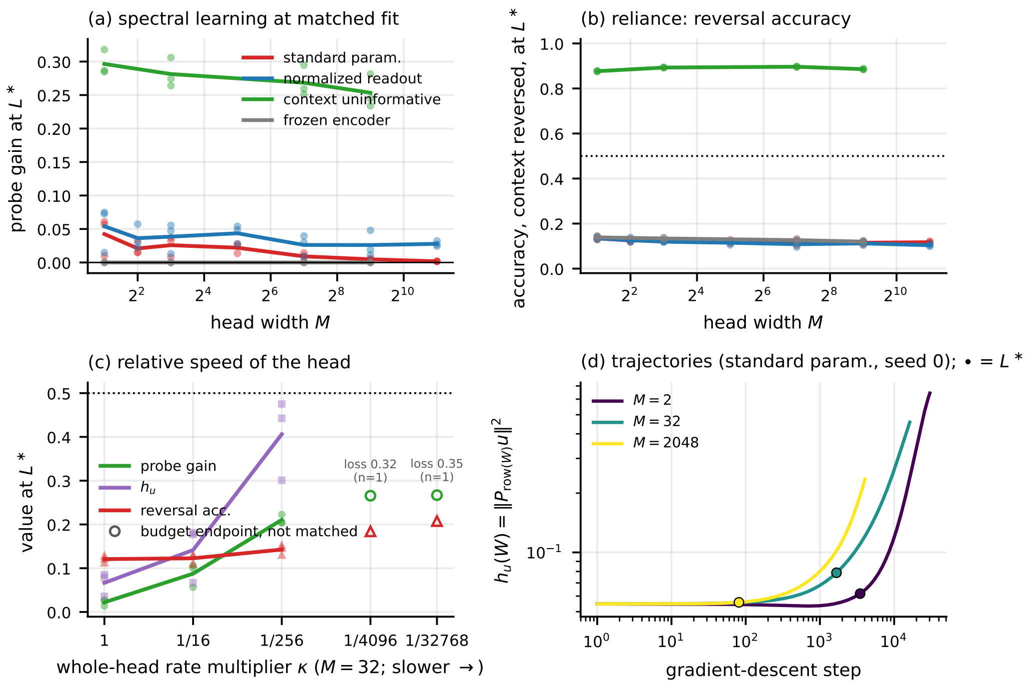

Figure 3: Experiment 2 at matched fit $L^* = 0.30$ (points are seeds, lines means). (a) Spectral probe gain against head width by arm. (b) Accuracy under context reversal. (c) Whole-head rate multiplier κ at $M = 32$: probe gain, h_u and reversal accuracy; open markers are single-seed budget endpoints of runs that did not reach L^* (losses shown), not matched-fit points. (d) h_u along training for three widths, L^* crossing marked.

original fit thus changed what retraining could recover by about 0.2; no retraining reached the 0.10 loss target, so this concerns the specified procedure and budget, not every possible repair.

Amendments. Widths 2 and 4, the fractional readout multipliers and the whole-head arm were added after a one-seed pilot and the design review, before their runs; the κ extension below $1/256$ was motivated by the observed reversal outcome; the recovery procedure was specified after the first results and before its runs (Appendix C).

5 Real centre spectra with constructed context

What this experiment is. We test contextual competition using measured centre spectra and their recorded labels, with a synthetic class-associated spatial intensity pattern along training-estimated principal directions in the neighbourhood. It asks whether the mechanism operates when the spectral cue is a real, high-dimensional class contrast; natural neighbourhoods are treated in Appendix E.

Data and readiness. Breast-tissue QCL micro-array cores, fold 0 of a five-fold patient-level split; the binary task Cancer Epithelium versus Cancer-Associated Stroma on 942-channel pixel vectors; $24{,}000$ class-balanced training centres; evaluation on held-out validation patients, with test patients reported separately (Appendix D). A readiness scan (Table 11) shows that with per-feature standardization a random encoder $W \in \mathbb{R}^{12\times 942}$ already reads this contrast at 0.955 against a calibration discriminant of 0.969, because the class-mean contrast lies along the top principal directions; PCA whitening estimated on the training patients reduces the random-encoder probe to 0.54 while the discriminant stays at 0.944. We therefore run two regimes on the same pixels and labels, *high* and *low initial probe accessibility* (standardized and whitened inputs). Neither is an instance of the theorem: a probe on the encoder output does not measure the spectral contribution to a random head's margin.

Construction. Each patch is a real centre with eight real donor spectra drawn independently of the centre label from the same split side; donor d carries $\gamma_d(c\,s + \tau\eta_d)$ along the dth principal coordinate, where c is the centre's label, $s = \pm 1$ or c is replaced by an independent label, and γ_d is large relative to the donors' natural variation ($3\sqrt{\lambda_d}$ standardized; 30 in whitened units, with a $\gamma = 10$ variant declared before its runs). τ is calibrated per regime so that a discriminant on the eight cue projections matches the centre-only discriminant on training-side data (0.969 high, 0.948 low); through a random encoder the encoded patch, centre included, reads at 0.94–0.99. The model is the linear encoder over the nine spectra and a ReLU head of width M over the encoded patch, trained and thresholded as in Section 4; three seeds per arm.

Results: high accessibility. The probe is 0.91–0.94 before training and changes little at L^* in any arm (gain -0.015 to $+0.035$), yet the informative-context-trained heads are strongly context-reliant, with reversal accuracy 0.08–0.10 and context-random accuracy at chance for widths 8–512 and for $\kappa = 1,\ 1/16,\ 1/256$ alike, all fitting within 70–230 steps; the uninformative-context comparator reaches reversal 0.93 and context-random 0.92, the frozen encoder 0.09 (Table 15). A large deficit in this probe is not necessary for failure.

Results: low accessibility. The probe starts at 0.56–0.60. Among the $\gamma = 30$ informative-context arms the mean gain at L^* is at most 0.046, with substantial seed variation (up to 0.10 in one run), against 0.35 for the uninformative-context comparator (probe 0.92–0.93; 15,000–17,000 steps to L^* against 59–226); reversal accuracy is 0.07 against the comparator's 0.91–0.93, the frozen encoder giving 0.08 (Table 13). Slowing the head gives only a small probe response (mean $+0.014$ from $\kappa = 1$ to $1/256$). The saved checkpoints show where the adaptation went: for every seed and κ, 99.995–99.998% of the squared net encoder update lies in the span of the eight constructed-cue directions, against 88–90% under uninformative-context training (Appendix D). That the cue's amplitude dominates the encoder's own gradient is a supported hypothesis, not an identified mechanism: the projection is not a pathwise gradient decomposition, and the random head, unlike the theorem's zero-sum readout, lets contextual learning proceed through the encoder even when the head barely moves.

Results: the lower-amplitude variant. At $\gamma = 10$ the fit takes 530–1,800 steps and the rate response returns: the within-seed probe change from $\kappa = 1$ to $1/256$ is $+0.030$, $+0.049$, $+0.095$, against $+0.002$, $+0.023$, $+0.016$ at $\gamma = 30$ in the same seeds (paired difference positive in every seed), while reversal accuracy stays at 0.06–0.08 and the comparator again reaches a gain of 0.35 and reversal 0.92 (Table 14). The net update remains in the cue span (99.96–99.98%): an amplitude-dependent accessibility response, not the removal of contextual dominance.

Results: what retraining recovers. The recovery procedure of Section 4, applied to these encoders, separates the regimes further (Table 16). With high accessibility a fresh head retrained on decorrelated context reaches reversal accuracy 0.90–0.96 from every encoder, the random initialization included, against 0.08 for the original classifiers: the spectral information was available and unused. At $\gamma = 30$ no encoder yields more than 0.45–0.56; at $\gamma = 10$ the slow-head encoders yield 0.55–0.70 against 0.48–0.60 after the fast head.

Reading. With measured spectra the outcome matches the synthetic experiment on reliance: at every width and rate evaluated the original predictor uses the constructed context when it is informative, with excess shifted risk of 0.83–0.84 over the uninformative-context comparator. On adaptation it adds a scope condition: how much of the contrast the encoder learns before the fit depends on the head's relative speed and on the cue's amplitude, which are not interchangeable here. The construction matters: with the natural 3×3 neighbourhoods of the same cores, which never straddle two classes, the encoder learns the class signal equally with and without informative context and the model uses the full spectrum, on tissue and on a public hyperspectral scene (Appendices E and F).

6 Limitations and Conclusion

Conclusion. We prove, in a serial linear-encoder model, that a contextual pathway trained faster than the spectral encoder suppresses the encoder's update at matched fit, that the fitted model fails

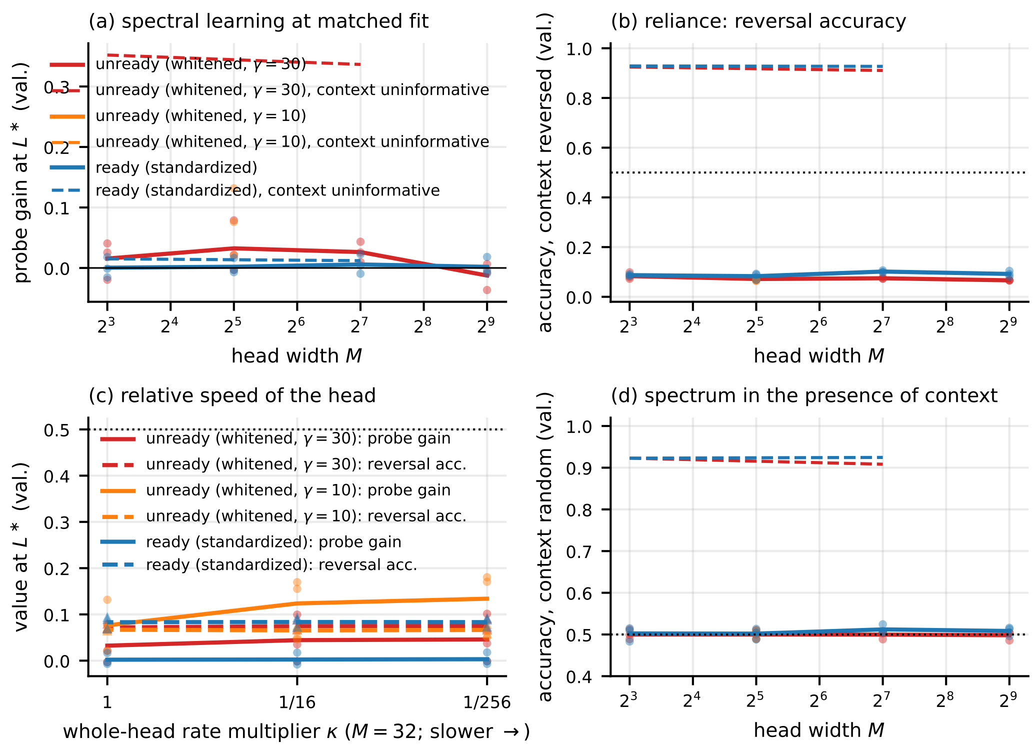

Figure 4: Experiment 3 at $L^* = 0.30$ on validation patients, three seeds per cell. (a) Probe gain against width: high accessibility (blue), low accessibility ($\gamma = 30$ red, $\gamma = 10$ orange), uninformative-context comparator dashed. (b) Accuracy under context reversal. (c) Whole-head rate multiplier at $M = 32$: probe gain (solid), reversal accuracy (dashed). (d) Accuracy with an uninformative context.

under contextual reversal under explicit finite-width conditions, and that readout normalization removes the width dependence without necessarily removing the failure. At matched training loss in the synthetic spectral–spatial model, slowing the spatial head raises the linear accessibility of the spectral cue while the trained classifier remains strongly context-reliant; retraining a fresh head on decorrelated context recovers about 0.2 more shifted accuracy after the slow head than after the fast one, which yields no more than a random encoder. With measured spectra, high spectral accessibility coexists with severe contextual reliance, so a deficit in this probe is not necessary for failure; with low initial accessibility the head's rate matters more at the smaller constructed amplitude. The paper's claim is that difference, at the stated fits, rates and constructions, not a general law.

Limitations. The theorem has a linear encoder, noiseless cues and a fixed spectral skip coefficient; the experiments add noisy cues and jointly learned routing but keep a linear encoder and one head family each. The cues that show the mechanism are constructed; natural 3×3 neighbourhoods, being the same class as their centre, did not produce it, and larger receptive fields are untested. Matched mean training loss is an analogue of the theorem's matched margin, not the same rule, and where a random encoder already reads the spectral cue the failure that still occurs is not the theorem's. Real-spectrum seed variation is conditional on one patient split. On a production segmentation model, scalar initialization curvature is not the diagnostic used here, and an earlier apparent freezing advantage was protocol-confounded (Appendix G). We test no mitigation on natural context; the supported statement is conditional: where the spectral cue is accessible at the encoder output, retraining the readout on context-decorrelated data recovered it; where it is not, only retraining the encoder without informative context did.

AI use statement

Generative AI tools (large language model assistants) were used throughout this work: for literature search; for drafting and editing text; for writing and checking code, including the numerical verification scripts of Appendix H; and for adversarial review of the proofs and of the claims made about the measurements. Every proof was independently re-derived by the authors, and every number reported in the paper was recomputed by the authors from the recorded artifacts. All AI-generated text and code were reviewed by the authors, who take full responsibility for the content of this paper, including any remaining errors.

Reproducibility statement

Code, configurations and audit scripts, with seeds and exact commands, are provided in the anonymized repository accompanying this submission. Every number in the paper traces to a named results file; Appendix H lists the verification scripts and their outcomes, including the two that do not exit cleanly. Complete hypotheses and proofs are in Appendix A, with the numerical verification in Appendix B. Generators, calibration, architectures, data, patient splits and training protocols are described in Appendices C, D and G.

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A PROOFS OF THEOREM 1

Theorem 1 combines three statements proved below in their original form: the serial instance at the special initialization (Theorem A.1), its general and isotropic initialization (Corollary A.1), and the readout normalization (Proposition A.2). Part (a) of Theorem 1 is (10) together with the finite-horizon gap (12); part (b) is the reversal statement of Corollary A.1 including (13); part (c) is Proposition A.2 with $\alpha_M = M^{-1/2}$, whose proof applies to any zero-sum initialization of the normalized readouts because only the aggregate b enters the flow. The rate-κ variant quoted in Section 3 follows by the same differentiation with $\dot{\beta}_j = \kappa v \varrho$. The special-initialization remark is (4) and the GGN display of Theorem A.1. Notation: $\varrho = (1 + e^q)^{-1}$ is the logistic residual, $\Psi(q) = q + e^q - 1$, and $G_{\boldsymbol{\theta\theta}}$, $G_{\boldsymbol{\phi\phi}}$ are the Gauss–Newton blocks of the scalar-logit loss in the parameter blocks $\boldsymbol{\theta} = (a, v)$ and $\boldsymbol{\phi} = \beta$.

A.0.1 THE SERIAL INSTANCE, ITS GENERAL INITIALIZATION, AND THE NORMALIZED READOUT

Proposition A.1 (Information retained by a fitting serial encoder)**.** *If the predictions depend on X only through $Z = f_{\boldsymbol{\theta}}(X)$ and a classifier $q_{\boldsymbol{\phi}}(Y \mid Z)$ has population CE at most $\eta < \infty$, then $H(Y \mid Z) \leq \eta$ and $I(Y;Z) \geq H(Y) - \eta$. The same statement holds for the empirical distribution with a fixed model.*

Proof. CE is conditional entropy plus the nonnegative expected conditional KL divergence. Suppression of a selected coordinate or failure under a specified shift is therefore different from absence of all training-label information in the serial representation. ☐

Theorem A.1 (Serial CE instance; review Theorem 5.1)**.** *Let Y be uniform on $\{-1, 1\}$ and let training inputs be $x_1 = (S, 0)$, $x_2 = (0, C_{\text{ctx}})$ with $S = C_{\text{ctx}} = Y$. The shared linear encoder is $W = (a, v)$, giving $z_1 = aS, z_2 = vC_{\text{ctx}}$. For integer $M \geq 1$ define*

$$F = z_1 + \sum_{j=1}^{M} \beta_j z_2 = aS + bvC_{\text{ctx}}, \quad b = \sum_j \beta_j, \quad \boldsymbol{\theta} = (a, v), \quad \boldsymbol{\phi} = \beta.$$

All coordinates follow unit Euclidean gradient flow on the averaged binary CE $\mathcal{L} = \mathbb{E} \log(1 + e^{-YF}) = \log(1 + e^{-q})$, $q = a + bv$, from $a_0 = 0, v_0 = 1, \beta_0 = 0$, with no clipping, decay or momentum. Fix $0 < \delta < 1/2$, $m = \log((1 - \delta)/\delta)$, and let T_m be the first time q reaches m. Write $\varrho = (1 + e^q)^{-1}$ and $\Psi(q) = q + e^q - 1$. Then:

$$\dot{a} = \varrho, \qquad \dot{v} = b\varrho, \qquad \dot{b} = Mv\varrho, \tag{1}$$

$$v = \cosh(\sqrt{M}a), \qquad b = \sqrt{M} \sinh(\sqrt{M}a), \tag{2}$$

$$q = a + \frac{\sqrt{M}}{2} \sinh(2\sqrt{M}a), \qquad \dot{q} = (M + 1 + 2b^2)\varrho. \tag{3}$$

The solution is global and reaches m in finite time, with

$$\frac{\Psi(m)}{M + 1 + 2m^2} \leq T_m \leq \frac{\Psi(m)}{M + 1}. \tag{4}$$

At that time, writing $a_M = a(T_m)$,

$$0 < a_M \leq \frac{m}{M + 1}, \qquad b(T_m)v(T_m) \geq \frac{Mm}{M + 1}, \qquad Ma_M \longrightarrow m, \tag{5}$$

$$0 \leq v(T_m) - 1 \leq \frac{m^2}{2M}, \qquad \|W(T_m) - W_0\| \leq \sqrt{\frac{m^2}{(M + 1)^2} + \frac{m^4}{4M^2}}. \tag{6}$$

For fixed m, a_M decreases strictly with M; removing the contextual path entirely requires $a = m$ at the same fitting margin. Moreover,

$$\varrho(t) \leq \frac{1/2}{1 + (M + 1)t/4}, \qquad t \geq 0. \tag{7}$$

In the stated parameter blocks and loss convention,

$$\lambda_{\max}(G_{\boldsymbol{\theta\theta}}) = \varrho(1 - \varrho)(1 + b^2), \quad \lambda_{\max}(G_{\boldsymbol{\phi\phi}}) = M\varrho(1 - \varrho)v^2,$$

so at initialization they equal $1/4, M/4$ and the disparity is M. For the separately trained frozen encoder $W = (0, 1)$, $\beta_F(0) = 0$, its margin obeys $\Psi(q_F(t)) = Mt$ and

$$0 \leq q(t) - q_F(t) \leq \frac{(1 + 2m^2)t}{2 + Mt/2} \leq \frac{2(1 + 2m^2)}{M}, \qquad 0 \leq t \leq T_m. \tag{8}$$

Under the specified reversal $S = Y, C_{\text{ctx}} = -Y$, the joint model at T_m has error one; the fitted spectral-only comparator has error zero.

Proof. The training margin is the same on each example, so direct differentiation gives (1). Divide by $\dot{a} = \varrho > 0$ to get $dv/da = b$, $db/da = Mv$, which proves (2). It implies $b^2 = M(v^2 - 1)$ and (3). Since $0 < \dot{a} \leq 1/2$, a stays finite on every finite time interval; the phase formula then gives global existence. The strictly increasing map $Q_M(a) = a + (\sqrt{M}/2)\sinh(2\sqrt{M}a)$ maps $[0, \infty)$ onto itself and has a finite inverse at m. On this compact a interval, $\dot{a} \geq \delta > 0$, so the hitting time is finite.

$\sinh x \geq x$ gives $Q_M(a) \geq (M+1)a$, strictly for $a > 0$. Before fitting, $v \geq 1$, $0 \leq bv = q - a \leq m$, hence $b \leq m$. The invariant gives $v - 1 = b^2/[M(v+1)] \leq m^2/(2M)$. For fixed positive a the contextual term increases strictly with M; invert to get strict decrease of a_M. Its upper bound implies $\sqrt{M}a_M \to 0$, and the expansion $m = (M+1)a_M + O(M^2 a_M^3)$ gives $Ma_M \to m$.

$\dot{\varrho} = -(M + 1 + 2b^2)\varrho^2(1 - \varrho)$ and $\varrho \leq 1/2$ imply $(d/dt)(1/\varrho) \geq (M+1)/2$. Integrate from $\varrho(0) = 1/2$ for the envelope. Since $\Psi'(q) = 1/\varrho$, before fitting $\frac{d}{dt}\Psi(q) = M + 1 + 2b^2 \in [M+1, M+1+2m^2]$; integration proves (4). The margin gradients are $Y(1, b)$ and $Yv\mathbf{1}_M$; the scalar logistic Hessian is $\varrho(1-\varrho)$, giving the GGN.

For the frozen system $\dot{q}_F = M/(1+e^{q_F})$. Thus $0 \leq \Psi(q) - \Psi(q_F) = \int_0^t (1+2b^2)ds \leq (1+2m^2)t$. Convexity gives a lower bound $\Psi'(q_F)(q - q_F)$ for this difference. As $q_F \leq e^{q_F} - 1$, $Mt = \Psi(q_F) \leq 2(e^{q_F} - 1)$, so $\Psi'(q_F) \geq 2 + Mt/2$, proving (8). The shifted margin is $a - bv = 2a_M - m < 0$: for $M > 1$ use (5), and for $M = 1$ use the strict inequality $Q_1(a) > 2a$. The spectral-only shifted margin is m. □

Remark A.1 (Embedding, head, and initialization conventions). For logits $(F, 0)$, loss, residual norm and GGN agree, but the unprojected logit-Jacobian norm is $\sqrt{1+b^2}$; the symmetric embedding makes the norm-product bound exact. A freely trained dense two-row head is not the parameter block $\boldsymbol{\phi} = \beta$: its initial head eigenvalue is $M/2$, not $M/4$, while the encoder eigenvalue remains $1/4$ at the same initial function. Labels need not be balanced for the training ODE; balance is used in the information/noisy-accuracy statement below. The initial encoder orientation in Theorem A.1 is explicitly asymmetric; the next corollary removes that choice for W, not the routing/readout asymmetry.

Proposition A.2 (Readout normalization and representation interpretation)**.** *Replace the contextual readout by $\alpha_M \sum_j \gamma_j z_2$, $\gamma_j(0) = 0$, at unit rates. Its reduced equation for $b = \alpha_M \sum_j \gamma_j$ is $\dot{b} = M\alpha_M^2 v\varrho$. In particular $\alpha_M = M^{-1/2}$ gives exactly the $M = 1$ dynamics for every M. At every finite M in the original theorem, the noiseless spectral coordinate $a_M S$ still determines Y. If it is instead observed as $Z_S = a_M Y + \sigma\xi$, with $\xi \sim \mathcal{N}(0,1)$ independent, $\sigma > 0$ fixed, then Bayes accuracy is $\Phi(a_M/\sigma)$ and $I(Y; Z_S) \leq a_M^2/(2\sigma^2)$.*

Proof. Differentiate each readout and sum. The Gaussian likelihood-ratio test is thresholding at zero, giving the accuracy. For $Q = \mathcal{N}(0, \sigma^2)$, $\mathbb{E}_Y \operatorname{KL}(P_{Z_S|Y} \| Q) = a_M^2/(2\sigma^2) = I(Y; Z_S) + \operatorname{KL}(P_{Z_S} \| Q)$. This noise is a specified observation perturbation, not training noise. □

Corollary A.1 (General and isotropic encoder initialization)**.** *In the serial model let $m > 0$, $a_0 < m$, $v_0 \neq 0$, and allow any readout initialization with $\sum_j \beta_{j0} = 0$. Set $u = a - a_0$, $\Delta = m - a_0 > 0$. Then*

$$v = v_0\cosh(\sqrt{M}u), \quad b = \sqrt{M}v_0\sinh(\sqrt{M}u), \tag{9}$$

$$q = a_0 + u + \frac{\sqrt{M}v_0^2}{2}\sinh(2\sqrt{M}u),$$

$$0 < u_M \leq \frac{\Delta}{1 + Mv_0^2}, \qquad Mu_M \longrightarrow \frac{\Delta}{v_0^2}, \qquad \frac{a(T_m) - a_0}{m - a_0} \longrightarrow 0. \tag{10}$$

The fitting root decreases strictly with M and

$$|v(T_m) - v_0| \leq \frac{\Delta^2}{2M|v_0|^3}, \qquad \|W(T_m) - W_0\| \leq \sqrt{\frac{\Delta^2}{(1+Mv_0^2)^2} + \frac{\Delta^4}{4M^2|v_0|^6}}. \tag{11}$$

With $\Psi_{a_0}(q) = q - a_0 + e^q - e^{a_0}$,

$$\frac{\Psi_{a_0}(m)}{1 + Mv_0^2 + 2\Delta^2/v_0^2} \leq T_m \leq \frac{\Psi_{a_0}(m)}{1 + Mv_0^2}.$$

For a frozen encoder at (a_0, v_0) and the same initial readouts, $\Psi_{a_0}(q_F) = Mv_0^2 t$. If $p_0 = e^{a_0}/(1 + e^{a_0})$ and $C_0 = 1 + 2\Delta^2/v_0^2$, then

$$0 \le q_J - q_F \le \frac{C_0 t}{1 + e^{a_0} + p_0 M v_0^2 t} \le \frac{C_0}{p_0 M v_0^2}, \qquad 0 \le t \le T_m. \tag{12}$$

Under contextual reversal the fitted margin is $2a(T_m) - m$. For $a_0 < m/2$, error is one for all sufficiently large M, with sufficient condition $Mv_0^2 > m/(m - 2a_0)$. For $m/2 \le a_0 < m$, the reversed classifier is correct at every width. If $a_0 \ge m$, define fitting to stop immediately, in which case it is also correct.

If $(a_0, v_0) \sim \mathcal{N}(0, \sigma^2 I_2)$, $\sigma > 0$, then $v_0 \neq 0$ almost surely; use the preceding stopping convention. The expected reversal error satisfies

$$\lim_{M\to\infty} \mathbb{E}_{W_0} \operatorname{Err}_{\text{reversal}} = \Phi\left(\frac{m}{2\sigma}\right). \tag{13}$$

The same-threshold spectral-only comparator has zero reversal error.

Proof. Dividing the ODE by $\dot{a} > 0$ gives (9); only the initial readout sum enters. Each $\beta_i - \beta_j$ is conserved, so equal initial readouts are not needed for closure. Positivity of the contextual margin $bv = (\sqrt{M}v_0^2/2)\sinh(2\sqrt{M}u)$ and the sinh inequality prove (10); Taylor expansion at $\sqrt{M}u_M \to 0$ gives the limit. The invariant is $b^2 = M(v^2 - v_0^2)$; v keeps its sign and $|v| \ge |v_0|$. Since $bv \le \Delta$, $|b| \le \Delta/|v_0|$ and

$$|v - v_0| = \frac{b^2}{M(|v| + |v_0|)} \le \frac{\Delta^2}{2M|v_0|^3}.$$

This finite-M inequality proves (11). Now $\dot{q} = (1 + Mv_0^2 + 2b^2)\varrho$ and $b^2 \le \Delta^2/v_0^2$ on the fitting interval; integrate the transformed margin for the time bounds. Global existence follows as before from $\dot{u} \le 1$ and the phase formula; every finite m is hit.

The transformed joint/frozen difference lies between zero and $C_0 t$. Also $q_F - a_0 \le (e^{q_F} - e^{a_0})/e^{a_0}$ gives $\Psi'_{a_0}(q_F) \ge 1 + e^{a_0} + p_0 M v_0^2 t$. Convexity proves (12). For $a_0 < m/2$, use $a(T_m) \le a_0 + \Delta/(1 + Mv_0^2)$ to get the finite-M reversal condition. For $a_0 > m/2$ the classifier is correct at every fitted width, since $u_M > 0$ gives $2a(T_m) - m > 2a_0 - m > 0$; equality $a_0 = m/2$ also gives a positive margin. Immediate stopping covers $a_0 \ge m$. Almost surely the limiting error is $1_{\{a_0 < m/2\}}$, apart from the zero-probability boundary and $v_0 = 0$ events. Dominated convergence gives (13). □

Remark A.2 (Confidence dependence and limiting quantity). For $a_0 \neq 0$, $a(T_m)/a_0 \to 1$ but $a(T_m)/m \to a_0/m$; the vanishing quantity is the *update* fraction in (10). For $0 < \rho < 1$, a Gaussian tail and its maximum density give, with probability at least $1 - \rho$,

$$|a_0| \le \sigma\sqrt{2\log(4/\rho)}, \qquad |v_0| \ge \frac{\rho\sigma}{2}\sqrt{\pi/2}.$$

Substitute these bounds in the finite-M inequalities for uniform confidence bounds. The dependence is severe: for fixed $(\Delta, v_0) \in (0, \infty) \times (\mathbb{R} \setminus \{0\})$, expansion of the root yields

$$Mu_M = \frac{\Delta}{v_0^2}\left[1 - \frac{1}{Mv_0^2} - \frac{2\Delta^2}{3Mv_0^4} + O(M^{-2})\right].$$

Indeed substitute $u = A/M + B/M^2$ in the sinh equation and match its constant and M^{-1} coefficients. The expansion's remainder is pointwise, not already uniform in rho. Its small-$|v_0|$ crossover requires control of $\Delta^2/(Mv_0^4)$; at the confidence cutoff this has scale $\Delta^2\rho^{-4}\sigma^{-4}/M$, a fourth power of $1/\rho$, not a second. A rigorous uniform remainder can instead be bounded from the sinh series on this controlled argument. Do not present this asymptotic crossover as a universal necessary width.

Provenance and interpretation. Architecture- and initialization-dependent implicit bias is established in linear-network analyses (Yun et al., 2021; Moroshko et al., 2020); trajectory-level incremental learning is studied by Berthier (2023). Exact deep-linear dynamics (Saxe et al., 2014) and conservation of layer-norm differences (Du et al., 2018) supply the underlying tools. CE feature competition is central to Pezeshki et al. (2021). The serial example instantiates a known optimization-metric mechanism with finite-margin formulas, an encoder-update comparison and a specified reversal distribution. Replication is equivalent to a rate M on b, and normalization removes it; the function class does not grow. These particular calculations are not a claim of a new general implicit-bias mechanism, and finite-time or serial composition alone is not a novelty claim.

B Exact-model verification

The closed forms of Appendix A were checked against a full $(2 + M)$-parameter integration of the gradient flow. The tables below are reproduced from the verification record of the earlier draft (`code/experiments/toy_serial_ce.py`, `toy_serial_ce_isotropic.py`); the maximum relative discrepancy over the four endpoint quantities in the 525 non-trivial isotropic cases is 1.26×10^{-11}, the maximum trajectory-identity discrepancy is 1.5×10^{-10}, and no bound is violated in any case. The quantities and error conventions are those stated in each caption.

B.0.1 Solvable serial instance: numerical verification

Zero-context instance. Table 1 verifies the closed forms and the bounds of the solvable serial cross-entropy instance at $a_0 = 0$, $v_0 = 1$ and margin $m = \log 19 = 2.944$, from `results/toy_serial_ce.csv`. The solver column is a stiff ODE integration of the full $(2 + M)$-dimensional state; a separate explicit-Euler gradient-descent path, run at four step sizes, has observed convergence order 1.0000004 and agrees with an independent numpy analytic-gradient implementation to 0 relative error.

Table 1: Solvable serial instance ($a_0 = 0$, $v_0 = 1$, $m = \log 19$): closed form against ODE solver, from `results/toy_serial_ce.csv`. The displacement block compares the realised encoder displacement with the displacement bound of the earlier draft's framework, which is not part of this paper's theory and is retained for the record only. The relative discrepancies between closed form and solver for the endpoint quantities in this table are at most 1.2×10^{-13}; the trajectory-invariant check of the next table reaches 2.0×10^{-11}.

Quantity	$M = 16$	$M = 64$	$M = 256$	$M = 1024$
$\lambda_{\boldsymbol{\theta}}(0)$	0.25	0.25	0.25	0.25
$\lambda_{\boldsymbol{\phi}}(0)$	4.0	16.0	64.0	256.0
$\mathcal{D}_{\mathrm{curv}}(0)$	16.0	64.0	256.0	1024.0
$a(T_m)$	0.142325	0.042094	0.011216	0.002857
suppression m/a_M	20.688	69.949	262.53	1030.7
$M\, a_M$ (limit m)	2.2772	2.6940	2.8712	2.9253
T_m	0.90784	0.28680	0.078763	0.020252
T_m lower bound $\Psi(m)/(M{+}1{+}2m^2)$	0.60992	0.25437	0.076345	0.020094
T_m upper bound $\Psi(m)/(M{+}1)$	1.2320	0.32222	0.081496	0.020434
$\sup_{0\le t\le T_m}(q_J - q_F)$	0.38232	0.13813	0.039802	0.010376
uniform gap bound	11.297	2.9547	0.74729	0.18737
$(M{+}1)\times$ gap	6.4994	8.9787	10.229	10.635
$\lVert W(T_m) - W(0)\rVert_2$	0.219021	0.071050	0.019658	0.005064
displacement bound	1.370601	0.358465	0.090662	0.022732
bound / displacement	6.258	5.045	4.612	4.489
normalized-readout control: $a(T_m)$	1.026947	1.026947	1.026947	1.026947
normalized-readout control: $\lVert W(T_m){-}W(0)\rVert$	1.177117	1.177117	1.177117	1.177117
residual envelope excess on $[0, 20T_m]$	0	0	0	0
readout spread $\max_j \beta_j - \min_j \beta_j$ at T_m	0	0	$1.4 \cdot 10^{-17}$	0
invariant $q = Q_M(a)$, max rel. error	$2.0 \cdot 10^{-11}$	$6.7 \cdot 10^{-13}$	$1.5 \cdot 10^{-13}$	$6.8 \cdot 10^{-14}$

The bound stays within 4.5 to 6.3 times the realised displacement across this width range while both fall by more than a factor of 40. The normalized-readout control, in which the readout is $M^{-1/2}Uh$ with $O(1)$ entries trained at unit rate, removes the width dependence exactly: the encoder's movement is the $M = 1$ value at every width, to 1.7×10^{-14} relative error. The eigenvalue constants $\lambda_{\boldsymbol{\theta}}(0) = 1/4$ and $\lambda_{\boldsymbol{\phi}}(0) = M/4$ hold for the mean-over-examples binary cross-entropy with a unit-scaled scalar logit; summing instead of averaging, or rescaling the logit, multiplies both blocks equally and leaves $\mathcal{D}_{\mathrm{curv}}(0) = M$ unchanged.

Isotropic extension. Table 2 collects the verification of the general-initialization version, in which $a(0) = a_0$, $v(0) = v_0 \neq 0$ and the readouts start unequal but sum to exactly zero. The grid is $M \in$

$\{1, 4, 16, 64, 256, 1024, 4096\}$, $a_0 \in \{-1.5, -0.3, 0, 0.4, 1.7\}$, $v_0 \in \{-1.3, -0.2, 0.05, 0.2, 1.0\}$ and $m \in \{2.0, 2.9, 4.0\}$: 525 non-trivial cases plus 54 immediate-stop cases with $a_0 \geq m$, integrated with DOP853 at `rtol` 10^{-10} and `atol` 10^{-12} with a terminal event at $q = m$.

Table 2: Isotropic serial instance: verification summary and worst-case bound slacks over the whole grid, from `results/toy_serial_ce_isotropic_REPORT.md` §1–§2. Slacks are bound − value for upper bounds and value − bound for lower bounds; a negative entry would be a violation, and there are none. $\Delta = m - a_0$.

Quantity	Value
Max closed-form vs ODE relative error over u, $v(T_m)$, $b(T_m)$, T_m (525 cases)	1.260×10^{-11}
Max relative error of the conserved quantity along the whole trajectory	1.518×10^{-10}
Max root accuracy $\lvert Q_{\text{iso}}(u_M) - m\rvert$	3.553×10^{-15}
Max $\lvert \sum_j \beta_j(0)\rvert$ (zero-sum initialization is bitwise exact)	0
Max spread of $\beta_j(T_m) - \beta_j(0)$	7.772×10^{-16}
Max frozen-comparator ODE vs closed-form relative error	5.993×10^{-11}
Max $M^{-1/2}$-readout control vs its $M = 1$ value	1.335×10^{-12}
Bound violations, all cases and all bounds	0
Min slack, $u_M \leq \Delta/(1 + Mv_0^2)$	2.221×10^{-10}
Min slack, $u_M > 0$	4.333×10^{-5}
Min slack, $\lvert v(T_m) - v_0\rvert \leq \Delta^2/(2M\lvert v_0\rvert^3)$	1.493×10^{-9}
Min slack, encoder displacement bound	3.918×10^{-10}
Min slack, $T_m \leq \Psi_{a_0}(m)/(1 + Mv_0^2)$	1.748×10^{-9}
Min slack, $T_m \geq \Psi_{a_0}(m)/(1 + Mv_0^2 + 2\Delta^2/v_0^2)$	3.175×10^{-9}
Min slack, $q_J(t) \geq q_F(t)$	0
Min slack, $\sup_t(q_J - q_F) \leq C_0/(p_0 M v_0^2)$	1.495×10^{-4}
Reversal error at $\sigma = 1$, $m = 2$, $M = 64$ / 1024 / 16384	0.8207 / 0.8380 / 0.8397
Monte-Carlo limit $\Phi(m/2\sigma)$ on the same 4000 draws	0.8413
Draws disagreeing with the limit that satisfy $Mv_0^2 > m/(m - 2a_0)$	0 of 48000
Spectral-only comparator reversal error, every case	0

The update fraction $(a(T_m) - a_0)/(m - a_0)$ falls towards zero in M at every (a_0, v_0) on the grid — for instance 0.288, 0.0423, 0.000964 at $M = 1, 16, 1024$ with $a_0 = -1.5$, $v_0 = 1$, $m = 2.9$ — while $a(T_m)/m$ and $a(T_m)/a_0$ converge to a_0/m and to 1 instead, so those two ratios are not the quantities that vanish. Under the $M^{-1/2}$ readout the suppression is 0.336399 at every width, against 0.336, 0.0472 and 0.000969 for the unnormalized readout at $M = 1$, 16 and 1024. Every Monte-Carlo draw that falls short of the reversal limit fails the sufficient finite-width condition, either because $\lvert v_0\rvert$ is small or because a_0 sits just below $m/2$, where the threshold diverges.

C Experiment 2 details

C.1 Generator

Images are 16×16 on a torus with $S = 256$ channels per pixel. Labels $y_p \in \{-1, +1\}$ are i.i.d. Rademacher. Fixed orthonormal directions $u, v_1, \ldots, v_8 \in \mathbb{R}^S$ are drawn once per seed (QR of a Gaussian matrix). With $o_1, \ldots, o_8$ the eight 3×3 offsets, the spectrum of pixel p is

$$x_p = \alpha\, y_p\, u + \beta \sum_{d=1}^{8} \big(y_{p-o_d} + \tau\, \eta_{p,d}\big) v_d + \sigma\, \varepsilon_p, \qquad \eta_{p,d}, \varepsilon_p \text{ standard Gaussian,}$$

so that the pixel at $p + o_d$ carries y_p along v_d: each pixel's label is written into all eight neighbours, each along its own direction, while p's own v-coordinates carry the labels of p's neighbours, which are independent of y_p. Constants: $\alpha = 1.645$, $\beta = 5$, $\sigma = 1$, $\tau = 1.7023$; $K = 12$. The draw order is condition-independent, so the five conditions generated from one seed share y, η and ε: *iid*; *reversed* ($y_{p-o_d} \to -y_{p-o_d}$); *context-random* (an independent label field replaces y in the context term); *spectral-only* ($\beta = 0$); *context-only* ($\alpha = 0$). Training uses 256 images; each test condition 64 images; the spectral probe is fitted on 64 spectral-only images and evaluated on another 64. Seeds:

directions $1000+s$, training data $2000+s$, test family $3000+s$, probe $4000+s$ and $5000+s$, initialization s, for $s \in \{0,1,2\}$.

C.2 Calibration and readiness (fixed before any training run)

The oracle spectral accuracy is a Fisher discriminant on $u^\top x_p$; the oracle contextual accuracy is a Fisher discriminant on the 3×3 window of the eight projections $v_d^\top x$ (72 features), which contains all eight copies of y_p. τ was set by bisection so that the contextual oracle equals 0.950 on seed-0 directions. Readiness is the same discriminant through a random encoder $W \sim \mathcal{N}(0, 1/S)^{12\times 256}$: on the encoder output of the pixel itself for spectral-only data, and on the 3×3 window of encoder outputs for context-only data.

Table 3: Experiment 2 calibration (`results/exp2/calibration.json`). Oracle accuracies use the true directions; readiness uses a random encoder.

seed	oracle spectral	oracle context	random enc. spectral	random enc. context	random enc. both
0	0.951	0.950	0.647	0.910	0.915
1	0.951	0.949	0.653	0.914	0.918
2	0.948	0.950	0.608	0.905	0.906

The population optimum of the contextual cue alone is a cross-entropy of about 0.126 nats (spectral alone 0.127); the threshold 0.15 sits just above that population optimum. A population optimum is not a bound on the finite-sample training loss, which a wide head can drive lower by fitting individual pixels, so training losses below 0.13 are not by themselves evidence of spectral learning; the probe and the shifted accuracies are.

C.3 Model, training and arms

Encoder $W \in \mathbb{R}^{12\times 256}$, no bias, $\mathcal{N}(0, 1/S)$ initialization. Head: 3×3 convolution to width M with bias, ReLU, 3×3 convolution to one logit without bias, circular padding, PyTorch default initialization; implemented as explicit matrix products with torus rolls (identical to the convolution to 10^{-15} in double precision). Loss: mean over the 65,536 training pixels of $\log(1+e^{-y_p F_p})$. Full-batch gradient descent, $\eta = 10^{-3}$ for every parameter (the largest of $\{10^{-4}, 3\cdot 10^{-4}, 10^{-3}, 3\cdot 10^{-3}\}$ with a monotone loss over 300 steps at $M = 2048$), no momentum, clipping or decay; at most 40,000 steps, stopping at loss < 0.10. Snapshots at the first step with loss below each of $0.6, 0.5, 0.4, 0.3, 0.2, 0.15, 0.10$; $L^* = 0.30$ is the pre-registered threshold. Gradient norms at a snapshot are those of the snapshotted iterate. Arms: `sp` (standard parameterization) and `mup` (readout $M^{-1/2}\gamma$, $\gamma = \sqrt{M}\times$default, same function at initialization; under plain gradient descent equal to a readout-only rate $1/M$) at $M \in \{2, 4, 8, 32, 128, 512, 2048\}$; `ctxfree` (training context from an independent label field) and `frozen` (encoder fixed at initialization) at $M \in \{2, 8, 128, 512\}$; `lrmult` (readout rate $\times\{1/16, 1/4, 1, 4, 16\}$) and `headlr` (whole-head rate $\kappa \in \{1, 1/16, 1/256\}$, extended to $1/4096$ and $1/32768$ on seed 0 for the reversal outcome) at $M = 32$. Three seeds.

C.4 Pre-registration and amendments

The plan (`paper/REFOCUS_PLAN_2026-09-10.md`, §4) fixed the generator, constants, arms, L^*, measures and predictions P1–P5 before the grid. Amendments recorded before the corresponding runs: widths 2 and 4 and fractional readout multipliers were added after a one-seed pilot at $M \in \{8, 2048\}$ showed complete suppression at L^* (§4.4a); the whole-head arm, the exact h_u export, paired test conditions and same-state gradient norms were added after the design review (§4.4b); the κ bracket was extended downward after the seed-0 pilot showed that the probe responds to κ while reversal accuracy does not within $\{1, 1/16, 1/256\}$ (§4.4c). Original predictions are reported as registered; added widths and multipliers are identified as amendments.

C.5 RESULTS

Tables generated by `paper/figures_src/make_tables_exp2.py` from `results/exp2_summary.csv`; the registered verdicts are the per-seed Spearman correlations listed at the end.

Table 4: Experiment 2 at the pre-registered threshold $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain is the discriminant accuracy on the encoder output minus its initial value (0.647, 0.653, 0.608 for seeds 0–2); $h_u = \|P_{\mathrm{row}(W)} u\|^2$ (0.055 at initialization).

arm	M	probe gain	h_u	reversed acc.	context-random acc.	spectral-only acc.
sp	2	0.042±0.028 (3)	0.086±0.037 (3)	0.134±0.006 (3)	0.503±0.005 (3)	0.525±0.016 (3)
sp	4	0.021±0.010 (3)	0.065±0.028 (3)	0.125±0.008 (3)	0.508±0.007 (3)	0.525±0.021 (3)
sp	8	0.026±0.016 (3)	0.067±0.021 (3)	0.124±0.002 (3)	0.507±0.005 (3)	0.524±0.017 (3)
sp	32	0.022±0.007 (3)	0.067±0.027 (3)	0.121±0.008 (3)	0.508±0.008 (3)	0.531±0.022 (3)
sp	128	0.009±0.006 (3)	0.055±0.022 (3)	0.117±0.012 (3)	0.507±0.008 (3)	0.524±0.019 (3)
sp	512	0.005±0.001 (3)	0.051±0.018 (3)	0.115±0.007 (3)	0.508±0.006 (3)	0.525±0.022 (3)
sp	2048	0.002±0.000 (3)	0.049±0.017 (3)	0.117±0.005 (3)	0.507±0.006 (3)	0.522±0.017 (3)
mup	2	0.054±0.034 (3)	0.098±0.042 (3)	0.134±0.006 (3)	0.504±0.004 (3)	0.528±0.015 (3)
mup	4	0.036±0.019 (3)	0.079±0.037 (3)	0.126±0.011 (3)	0.507±0.007 (3)	0.530±0.026 (3)
mup	8	0.039±0.023 (3)	0.080±0.034 (3)	0.120±0.002 (3)	0.506±0.006 (3)	0.530±0.024 (3)
mup	32	0.044±0.014 (3)	0.088±0.037 (3)	0.115±0.009 (3)	0.507±0.007 (3)	0.539±0.026 (3)
mup	128	0.026±0.016 (3)	0.070±0.031 (3)	0.108±0.012 (3)	0.505±0.008 (3)	0.526±0.018 (3)
mup	512	0.026±0.019 (3)	0.071±0.034 (3)	0.112±0.008 (3)	0.505±0.007 (3)	0.526±0.022 (3)
mup	2048	0.028±0.004 (3)	0.070±0.018 (3)	0.104±0.005 (3)	0.504±0.006 (3)	0.523±0.010 (3)
ctxfree	2	0.296±0.019 (3)	0.863±0.048 (3)	0.877±0.002 (3)	0.876±0.003 (3)	0.877±0.002 (3)
ctxfree	8	0.281±0.022 (3)	0.741±0.014 (3)	0.893±0.001 (3)	0.893±0.001 (3)	0.899±0.002 (3)
ctxfree	128	0.268±0.023 (3)	0.650±0.014 (3)	0.896±0.001 (3)	0.895±0.002 (3)	0.903±0.004 (3)
ctxfree	512	0.253±0.025 (3)	0.564±0.005 (3)	0.886±0.002 (3)	0.887±0.004 (3)	0.891±0.004 (3)
frozen	2	0.000±0.000 (3)	0.048±0.017 (3)	0.139±0.009 (3)	0.508±0.006 (3)	0.521±0.018 (3)
frozen	8	0.000±0.000 (3)	0.048±0.017 (3)	0.134±0.006 (3)	0.511±0.007 (3)	0.524±0.019 (3)
frozen	128	0.000±0.000 (3)	0.048±0.017 (3)	0.127±0.008 (3)	0.508±0.008 (3)	0.524±0.021 (3)
frozen	512	0.000±0.000 (3)	0.048±0.017 (3)	0.120±0.006 (3)	0.508±0.008 (3)	0.526±0.019 (3)

Table 5: Experiment 2 at the secondary threshold 0.15 (labelled; not pre-registered): mean±sd over seeds that reached it.

arm	M	probe gain	h_u	reversed acc.	context-random acc.
sp	2	0.219±0.039 (3)	0.462±0.232 (3)	0.131±0.088 (3)	0.543±0.046 (3)
sp	4	0.094±0.019 (3)	0.149±0.056 (3)	0.070±0.006 (3)	0.506±0.007 (3)
sp	8	0.121±0.020 (3)	0.184±0.046 (3)	0.073±0.007 (3)	0.508±0.008 (3)
sp	32	0.094±0.014 (3)	0.147±0.051 (3)	0.072±0.008 (3)	0.509±0.009 (3)
sp	128	0.060±0.022 (3)	0.106±0.052 (3)	0.070±0.009 (3)	0.507±0.009 (3)
sp	512	0.041±0.012 (3)	0.084±0.036 (3)	0.071±0.011 (3)	0.508±0.009 (3)
sp	2048	0.032±0.005 (3)	0.073±0.026 (3)	0.076±0.010 (3)	0.509±0.010 (3)
mup	2	0.230±0.032 (3)	0.499±0.216 (3)	0.136±0.093 (3)	0.546±0.051 (3)
mup	4	0.142±0.020 (3)	0.233±0.079 (3)	0.074±0.008 (3)	0.508±0.006 (3)
mup	8	0.148±0.030 (3)	0.241±0.090 (3)	0.073±0.009 (3)	0.509±0.008 (3)
mup	32	0.134±0.015 (3)	0.213±0.068 (3)	0.070±0.008 (3)	0.507±0.008 (3)
mup	128	0.088±0.033 (3)	0.146±0.070 (3)	0.064±0.006 (3)	0.505±0.008 (3)
mup	512	0.108±0.041 (3)	0.178±0.104 (3)	0.067±0.009 (3)	0.506±0.007 (3)
mup	2048	0.100±0.006 (3)	0.153±0.035 (3)	0.065±0.003 (3)	0.506±0.007 (3)
ctxfree	2	0.317±0.024 (2)	0.990±0.000 (2)	0.940±0.006 (2)	0.940±0.007 (2)
ctxfree	8	0.308±0.023 (3)	0.977±0.005 (3)	0.937±0.002 (3)	0.938±0.002 (3)
ctxfree	128	0.304±0.023 (3)	0.937±0.002 (3)	0.941±0.000 (3)	0.941±0.000 (3)
ctxfree	512	0.302±0.023 (3)	0.913±0.004 (3)	0.940±0.001 (3)	0.940±0.000 (3)

Runs that did not reach L^*. The following runs ended at the step budget above L^*: `headlr` $M = 32$, $\kappa = 0.000244141$, seed 0 (final loss 0.320 at step 40000); `headlr` $M = 32$, $\kappa = 3.05176e-05$, seed 0 (final loss 0.347 at step 40000).

Table 6: Experiment 2 at the secondary threshold 0.1 (labelled; not pre-registered): mean±sd over seeds that reached it.

arm	M	probe gain	h_u	reversed acc.	context-random acc.
`sp`	2	0.274±0.021 (2)	0.629±0.021 (2)	0.137±0.006 (2)	0.555±0.008 (2)
`sp`	4	0.227±0.006 (3)	0.462±0.055 (3)	0.089±0.009 (3)	0.525±0.007 (3)
`sp`	8	0.241±0.024 (3)	0.514±0.028 (3)	0.099±0.008 (3)	0.531±0.008 (3)
`sp`	32	0.227±0.016 (3)	0.459±0.044 (3)	0.098±0.008 (3)	0.530±0.007 (3)
`sp`	128	0.206±0.011 (3)	0.383±0.044 (3)	0.096±0.010 (3)	0.528±0.009 (3)
`sp`	512	0.170±0.006 (3)	0.282±0.047 (3)	0.096±0.010 (3)	0.528±0.009 (3)
`sp`	2048	0.142±0.007 (3)	0.218±0.038 (3)	0.106±0.013 (3)	0.533±0.011 (3)
`mup`	2	0.281±0.023 (2)	0.669±0.017 (2)	0.138±0.003 (2)	0.555±0.006 (2)
`mup`	4	0.250±0.008 (3)	0.566±0.063 (3)	0.095±0.011 (3)	0.529±0.008 (3)
`mup`	8	0.248±0.003 (2)	0.618±0.051 (2)	0.102±0.014 (2)	0.536±0.009 (2)
`mup`	32	0.243±0.002 (2)	0.591±0.024 (2)	0.089±0.000 (2)	0.528±0.001 (2)
`mup`	128	0.237 (1)	0.579 (1)	0.084 (1)	0.524 (1)
`mup`	512	0.242 (1)	0.611 (1)	0.086 (1)	0.525 (1)

Table 7: Whole-head rate multiplier κ at $M = 32$ (arm `headlr`) at L^*; rows marked † never reached L^* within 40,000 steps and show the final state (loss in parentheses).

seed	κ	step	probe gain	h_u	reversed	context-random	spectral-only
0	3.05176e-05†	40000 (0.347)	0.267	0.701	0.208	0.528	0.689
0	0.000244141†	40000 (0.320)	0.266	0.693	0.184	0.529	0.697
0	0.00390625	22708	0.223	0.475	0.152	0.519	0.641
0	0.0625	7611	0.106	0.177	0.129	0.513	0.568
0	1	1671	0.028	0.079	0.128	0.515	0.536
1	0.00390625	34174	0.203	0.443	0.146	0.510	0.615
1	0.0625	8977	0.100	0.181	0.129	0.509	0.575
1	1	1580	0.025	0.085	0.121	0.511	0.551
2	0.00390625	32068	0.205	0.301	0.130	0.500	0.570
2	0.0625	8704	0.056	0.067	0.110	0.498	0.520
2	1	1881	0.014	0.036	0.112	0.500	0.507

Table 8: Readout learning-rate multiplier at $M = 32$ (arm `lrmult`) at L^*: mean±sd over three seeds.

multiplier	probe gain	h_u	reversed acc.	context-random acc.
0.0625	0.041±0.013	0.085±0.036	0.115±0.009	0.507±0.007
0.25	0.031±0.012	0.076±0.033	0.116±0.009	0.507±0.007
1	0.022±0.007	0.067±0.027	0.121±0.008	0.508±0.008
4	0.015±0.003	0.059±0.022	0.127±0.008	0.509±0.008
16	0.008±0.001	0.054±0.019	0.133±0.009	0.510±0.006

Table 9: Recovery at $M = 32$: a fresh head with paired initialization trained for 20,000 steps on newly sampled context-random images with the encoder frozen at its L^* checkpoint (`init` = the random initial encoder). Original = the run's own classifier at L^*; retrained = the new head; accuracies on the same paired test family.

seed	encoder	probe	retrain loss	orig. reversed	orig. ctx-random	retr. iid	retr. reversed	retr. ctx-random
0	`init`	0.647	0.658	–	–	0.592	0.599	0.596
0	`kappa1`	0.675	0.645	0.128	0.515	0.608	0.618	0.612
0	`kappa256`	0.870	0.328	0.152	0.519	0.845	0.840	0.844
1	`init`	0.658	0.659	–	–	0.596	0.590	0.593
1	`kappa1`	0.683	0.658	0.121	0.511	0.594	0.591	0.592
1	`kappa256`	0.861	0.447	0.146	0.510	0.776	0.771	0.773
2	`init`	0.615	0.682	–	–	0.562	0.533	0.546
2	`kappa1`	0.628	0.676	0.112	0.500	0.574	0.539	0.554
2	`kappa256`	0.820	0.470	0.130	0.500	0.757	0.750	0.755

Table 10: Registered predictions (REFOCUS_PLAN §4.4) evaluated at $L^* = 0.30$ on the original five widths and on the amended grid; ρ = Spearman rank correlation across widths (or multipliers), one value per seed. P2's second clause compares a_u at $M = 2048$.

criterion	grid	per-seed values	verdict
P1: standard param., $\rho(a_u, M) \leq -0.8$ and $\rho(\text{rev}, M) \leq -0.8$ in every seed	original	a_u: -0.70, -1.00, -1.00; rev: -0.60, -0.90, -0.60	not met (1/3 seeds)
	amended	a_u: -0.54, -1.00, -0.96; rev: -0.64, -0.96, -0.82	not met
P2a: normalized readout, $\lvert\rho(\cdot, M)\rvert < 0.5$ in at least two seeds	original	a_u: +0.10, -0.90, -0.30; probe: -0.20, -0.90, -0.30; rev: -0.90, -0.70, -0.70	alignment met; probe met; reversal not met
	amended	a_u: +0.25, -0.96, -0.54; rev: -0.89, -0.89, -0.89	alignment not met; reversal not met
P2b: $a_u(2048)$ normalized $>$ standard in every seed	paired seeds	0.0059 vs 0.0045; 0.0062 vs 0.0045; 0.0043 vs 0.0027	met
P3: uninformative context, $a_u > 0.25$ at every width and $\lvert\rho(a_u, M)\rvert < 0.5$	amended control widths $\{2, 8, 128, 512\}$	a_u range 0.058–0.174; $\rho(a_u, M)$: -1.00, -1.00, -1.00; $\rho(\text{probe}, M)$: -1.00, -1.00, -1.00	cutoff not met; flatness not met (probe 0.936→0.893)
P4: readout multiplier, a_u and reversal accuracy decrease with the multiplier in every seed	run multipliers $\{1/16, 1/4, 1, 4, 16\}$ (64 dropped)	a_u: -1.00, -1.00, -1.00; rev: +1.00, +1.00, +1.00	a_u met; reversal not met (opposite sign)
P5: frozen encoder, reversal accuracy < 0.5 at every width	amended widths	max 0.146	met
P6 (amendment 4.4b): whole-head κ, probe, h_u and reversal accuracy increase as κ decreases, $\rho \leq -0.8$ in every seed	$\kappa \in \{1, 1/16, 1/256\}$, three seeds	probe: -1.00, -1.00, -1.00; h_u: -1.00, -1.00, -1.00; rev: -1.00, -1.00, -0.50	probe and h_u met; reversal not met (2/3 seeds)

D Experiment 3 details

D.1 Data

Breast-tissue QCL micro-array cores, fold 0 of the five-fold patient-level split used throughout this work (115 training, 28 validation, 26 test cores). Each labelled tissue pixel contributes the 942-dimensional vector of absorbance, first and second derivative (3×314 channels), after the field-standard preprocessing of the companion pipeline, which includes per-core PCA denoising with 23 components before the derivatives are taken; each core's spectra are therefore of numerical rank 23, while the pooled training-patient data are not, since the retained subspaces differ between cores. Up to 600 (training) or 300 (validation, test) pixels per class per core were sampled with a fixed seed: 75,297 training pixels from 79 cores, 11,024 validation from 20, 8,038 test from 16, over the four classes; the experiment uses the pair Cancer Epithelium (+1; 29,603 training pixels) versus Cancer-Associated Stroma (−1; 21,923).

D.2 Readiness scan

For each class pair, bottleneck K and preprocessing, Table 11 reports the centre-only oracle (ridge Fisher discriminant on the preprocessed 942-vector, fitted on one half of the training cores and evaluated on the other half) and the same discriminant through three random encoders $W \sim \mathcal{N}(0, 1/942)^{K\times 942}$. Standardization is per feature; whitening is PCA whitening with a relative ridge of 10^{-3} on the eigenvalues; both are estimated on the fitting half.

Table 11: Readiness scan (`results/exp3/readiness_scan.csv`): centre-only oracle and random-encoder probe (mean over three encoders) by pair, K and preprocessing. $\lvert\cos\rvert$ is the cosine between the class-mean contrast and the first principal direction of the standardized training half.

pair	preprocessing	oracle	$K = 2$	$K = 4$	$K = 12$	$K = 32$	$\lvert\cos\rvert$
CancerEpi vs CAS	standardized	0.969	0.726	0.803	0.955	0.968	0.50
CancerEpi vs CAS	whitened	0.944	0.503	0.486	0.539	0.591	0.50
NormalStroma vs CAS	standardized	0.810	0.580	0.639	0.674	0.741	0.44
NormalStroma vs CAS	whitened	0.768	0.502	0.509	0.529	0.600	0.44
NormalEpi vs CancerEpi	standardized	0.758	0.547	0.573	0.662	0.724	0.20
NormalEpi vs CancerEpi	whitened	0.761	0.575	0.545	0.624	0.580	0.20
NormalStroma vs CancerEpi	standardized	0.989	0.781	0.874	0.978	0.990	0.88
NormalStroma vs CancerEpi	whitened	0.974	0.579	0.541	0.549	0.626	0.88

With standardized inputs a random encoder already reads the epithelium-versus-stroma contrasts, because they lie along the top principal directions; whitening removes that accessibility while the oracle stays high. The two regimes of Section 5 follow: the same pair, $K = 12$, standardized (*ready*) or whitened (*unready*).

D.3 Patch construction

Each training example is a real centre pixel with its recorded label and eight real donor spectra sampled uniformly from the same split side (resampled once if the donor's core equals the centre's), independent of the centre label. In the preprocessed space, donor d receives the constructed cue $\gamma_d\,(c\,s + \tau\eta_d)\,e_d$, where $e_1, \ldots, e_8$ are the first eight principal coordinates (in the standardized regime the directions $v_1, \ldots, v_8$ with $\gamma_d = 3\sqrt{\lambda_d}$; in the whitened regime the first eight whitened coordinates with $\gamma_d = 30$), c is the centre's label code, and $s = +1$ (informative), $s = -1$ (reversed) or c replaced by an independent random label (context-random); *spectral-only* removes the cue and keeps the donors. The draw order is condition-independent, so the four conditions generated from one seed share centres, donors and η. Preprocessing, principal directions and λ_d are estimated on training pixels only. Training: 24,000 class-balanced centres; probe fitting: 6,000 disjoint training pixels; evaluation: up to 6,000 validation and 6,000 test centres with donors from their own side. τ is calibrated per regime on training-core halves so that the neighbourhood oracle (a discriminant on the eight cue projections) matches the centre-only oracle.

Table 12: Experiment 3 calibration (`results/exp3/calibration3.json`). Readiness: discriminant through a random 12×942 encoder on the centre alone and on the encoded 3×3 patch (three encoders).

regime	τ	centre oracle (fit/val)	context oracle	random enc. centre	random enc. patch
ready (standardized)	1.475	0.969 / 0.959	0.969	0.935, 0.970, 0.952	0.988, 0.994, 0.991
unready (whitened)	1.735	0.948 / 0.884	0.948	0.476, 0.512, 0.605	0.948, 0.947, 0.950

D.4 Model, training and arms

Encoder $W \in \mathbb{R}^{12\times942}$ applied to the nine spectra; head $h = \mathrm{ReLU}(W_1\,\mathrm{vec}(z) + b_1)$ of width M over the $9K = 108$ encoded coordinates and a linear readout to one logit; margin loss; full-batch gradient descent with $\eta = 10^{-3}$, at most 40,000 steps, the same thresholds and snapshot rule as Experiment 2. Arms per regime: `sp` at $M \in \{8, 32, 128, 512\}$; `headlr` with $\kappa \in \{1, 1/16, 1/256\}$ at $M = 32$; `ctxfree` (context-random training) at $M \in \{8, 128\}$; `frozen` at $M = 32$; seeds 0–2. Spectral probe: ridge discriminant on the encoder output of centre pixels, fitted on the probe set and evaluated on validation (and test) centres, reported as the gain over its initial value. Reliance and failure are measured on validation patients with the paired conditions; test patients are reported separately and were not used for any choice.

D.5 Results

Tables generated by `paper/figures_src/make_tables_exp3.py` from `results/exp3_summary.csv`.

Runs that did not reach L^*. Every run reached L^*.

E Experiment 4 details: natural context

E.1 Purpose and registration

Experiment 4 tests, with natural neighbourhoods, the inference that motivated this work: that a spatial–spectral model's insensitivity to spectral compression shows the spectral dimension to be largely redundant (O'Leary et al., 2026; Müller et al., 2023). It is a usage test, not a mechanism test: no cue is constructed or calibrated. Its predictions, arms and disposition rules were fixed before

Table 13: Experiment 3, unready (whitened, $\gamma = 30$) regime, at $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
`sp`	8	126±51	0.015±0.031	0.083±0.014	0.500±0.012	0.485±0.028	0.083±0.005	0.496±0.000 (3)
`sp`	32	113±22	0.032±0.042	0.072±0.006	0.500±0.010	0.510±0.006	0.070±0.003	0.495±0.001 (3)
`sp`	128	100±12	0.026±0.017	0.075±0.003	0.500±0.010	0.495±0.015	0.073±0.006	0.496±0.004 (3)
`sp`	512	69±8	-0.012±0.022	0.066±0.001	0.498±0.011	0.486±0.010	0.067±0.005	0.495±0.004 (3)
`headlr`	32 / 0.00390625	185±43	0.046±0.052	0.075±0.012	0.500±0.013	0.512±0.011	0.072±0.002	0.494±0.001 (3)
`headlr`	32 / 0.0625	177±40	0.044±0.051	0.074±0.011	0.500±0.013	0.512±0.010	0.073±0.001	0.495±0.002 (3)
`headlr`	32	113±22	0.032±0.042	0.072±0.006	0.500±0.010	0.510±0.006	0.070±0.003	0.495±0.001 (3)
`ctxfree`	8	16780±2138	0.352±0.008	0.925±0.015	0.923±0.015	0.924±0.010	0.935±0.023	0.934±0.023 (3)
`ctxfree`	128	14954±488	0.336±0.014	0.911±0.030	0.908±0.029	0.926±0.028	0.930±0.014	0.928±0.009 (3)
`frozen`	32	395±95	0.000±0.000	0.076±0.006	0.500±0.009	0.515±0.008	0.076±0.002	0.496±0.002 (3)

Table 14: Experiment 3, unready (whitened, $\gamma = 10$; amendment) regime, at $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
`sp`	32	592±70	0.076±0.055	0.067±0.004	0.501±0.011	0.512±0.007	0.063±0.002	0.496±0.002 (3)
`headlr`	32 / 0.00390625	1579±318	0.134±0.072	0.066±0.010	0.501±0.011	0.508±0.015	0.065±0.005	0.496±0.001 (3)
`headlr`	32 / 0.0625	1337±227	0.124±0.068	0.065±0.009	0.501±0.011	0.508±0.013	0.065±0.003	0.496±0.001 (3)
`headlr`	32	592±70	0.076±0.055	0.067±0.004	0.501±0.011	0.512±0.007	0.063±0.002	0.496±0.002 (3)
`ctxfree`	32	16003±231	0.350±0.012	0.916±0.005	0.916±0.006	0.925±0.009	0.923±0.011	0.924±0.011 (3)
`frozen`	32	2006±310	0.000±0.000	0.080±0.008	0.502±0.011	0.511±0.007	0.073±0.004	0.495±0.001 (3)

Table 15: Experiment 3, ready (standardized) regime, at $L^* = 0.30$: mean±sd over seeds (number of seeds). Probe gain = discriminant accuracy on the encoder output of validation centres minus its initial value; accuracies on validation (val) and test (test) patients under the paired conditions.

arm	M / κ	step	probe gain	rev. (val)	ctx-rand. (val)	spec.-only (val)	rev. (test)	ctx-rand. (test)
`sp`	8	132±59	0.000±0.017	0.087±0.004	0.502±0.017	0.522±0.035	0.088±0.009	0.499±0.003 (3)
`sp`	32	118±22	0.002±0.013	0.083±0.013	0.502±0.013	0.526±0.026	0.088±0.013	0.500±0.003 (3)
`sp`	128	103±20	0.006±0.016	0.102±0.004	0.512±0.011	0.578±0.060	0.102±0.009	0.511±0.008 (3)
`sp`	512	70±3	0.002±0.014	0.092±0.012	0.508±0.011	0.544±0.017	0.092±0.021	0.504±0.005 (3)
`headlr`	32 / 0.00390625	193±31	0.003±0.013	0.083±0.010	0.503±0.013	0.525±0.028	0.086±0.008	0.501±0.001 (3)
`headlr`	32 / 0.0625	184±29	0.002±0.013	0.084±0.011	0.503±0.013	0.524±0.028	0.086±0.008	0.501±0.001 (3)
`headlr`	32	118±22	0.002±0.013	0.083±0.013	0.502±0.013	0.526±0.026	0.088±0.013	0.500±0.003 (3)
`ctxfree`	8	1128±116	0.015±0.021	0.929±0.002	0.923±0.001	0.931±0.005	0.953±0.008	0.947±0.004 (3)
`ctxfree`	128	841±112	0.012±0.021	0.927±0.006	0.925±0.006	0.929±0.004	0.955±0.001	0.954±0.000 (3)
`frozen`	32	442±114	0.000±0.000	0.091±0.022	0.504±0.013	0.529±0.021	0.093±0.027	0.499±0.006 (3)

Table 16: Recovery at $M = 32$ (real spectra): a fresh patch head with paired initialization trained for 20,000 steps on newly sampled context-random patches with the encoder frozen at its L^* checkpoint (`init` = random initial encoder); validation and test patients under the paired conditions; original = the run's own classifier at L^* (validation).

regime	seed	encoder	probe (val)	orig. rev. (val)	retr. iid (val)	retr. rev. (val)	retr. ctx-rand. (val)	retr. rev. (test)
ready	0	`init`	0.912	–	0.905	0.902	0.902	0.935
ready	0	`kappa1`	0.928	0.068	0.949	0.943	0.947	0.960
ready	0	`kappa256`	0.929	0.072	0.947	0.939	0.943	0.960
ready	1	`init`	0.941	–	0.951	0.953	0.952	0.969
ready	1	`kappa1`	0.934	0.093	0.958	0.959	0.958	0.974
ready	1	`kappa256`	0.934	0.090	0.955	0.959	0.957	0.973
ready	2	`init`	0.919	–	0.915	0.912	0.913	0.951
ready	2	`kappa1`	0.916	0.089	0.924	0.923	0.923	0.953
ready	2	`kappa256`	0.918	0.087	0.937	0.928	0.931	0.959
unready	0	`init`	0.585	–	0.588	0.445	0.522	0.440
unready	0	`kappa1`	0.582	0.068	0.591	0.447	0.526	0.460
unready	0	`kappa256`	0.584	0.066	0.591	0.458	0.527	0.493
unready	1	`init`	0.600	–	0.496	0.546	0.516	0.535
unready	1	`kappa1`	0.679	0.078	0.509	0.559	0.536	0.552
unready	1	`kappa256`	0.701	0.089	0.523	0.562	0.543	0.556
unready	2	`init`	0.560	–	0.522	0.523	0.527	0.546
unready	2	`kappa1`	0.582	0.068	0.517	0.549	0.543	0.564
unready	2	`kappa256`	0.598	0.070	0.527	0.557	0.547	0.571
unready10	0	`init`	0.585	–	0.596	0.431	0.517	0.427
unready10	0	`kappa1`	0.606	0.063	0.622	0.477	0.553	0.518
unready10	0	`kappa256`	0.636	0.057	0.638	0.550	0.598	0.626
unready10	1	`init`	0.600	–	0.504	0.556	0.528	0.556
unready10	1	`kappa1`	0.732	0.071	0.574	0.603	0.586	0.608
unready10	1	`kappa256`	0.780	0.077	0.684	0.700	0.692	0.716
unready10	2	`init`	0.560	–	0.527	0.532	0.535	0.546
unready10	2	`kappa1`	0.636	0.066	0.552	0.604	0.579	0.611
unready10	2	`kappa256`	0.731	0.065	0.653	0.670	0.664	0.689

the first run and are reproduced in the pre-registration file; two amendments were recorded before the arms they concern were run (the class-weighted loss, and the no-bottleneck arms added after the authors' code was checked). Every registered outcome is reported, including the ones that failed.

E.2 Data and natural patches

The cores, patient-level fold-0 split and four classes are those of Appendix D. Two copies of the spectra are used: the non-denoised copy (full rank; the pre-registered primary set) and the companion pipeline's PCA-23-denoised copy (the field-standard preprocessing, closer to the pipeline of O'Leary et al. (2026); a replication with identical arms and predictions, registered before its cache was built). A patch is a labelled centre pixel with its eight recorded neighbours, all nine inside the tissue mask; neighbours keep their own labels or lack of one. Per core and class, up to 150 (training) or 100 (validation, test) patches were sampled with a fixed seed: 19,650 training patches from 79 cores, 3,700 validation from 20, 2,700 test from 16 (54 cores carry no label of the four classes). The neighbourhoods are homogeneous: no patch contains a neighbour of a different labelled class, 88% of neighbours share the centre's label and 12% are unlabelled. Natural 3×3 context on this data is therefore a redundant copy of the centre's class signal, not a separate cue.

E.3 Stage A: per-pixel information against the number of components

On the standardized centre spectra, with PCA fitted on training patients, three per-pixel classifiers were fitted on training centres for each number of components k and scored on validation and test patients: a shrinkage Fisher discriminant, balanced logistic regression and a 256-unit ReLU MLP with a fixed 40-epoch schedule and three seeds, with no selection on the evaluation patients (Tables 17 and 21). The registered gate was a gain of at least 0.02 macro-F1 from $k = 16$ to the full spectrum on both evaluation sides. On the non-denoised copy the gains are 0.065–0.107 (LDA $0.560 \rightarrow 0.651$ validation, $0.538 \rightarrow 0.645$ test; logistic $0.595 \rightarrow 0.673$, $0.571 \rightarrow 0.658$; MLP $0.640 \rightarrow 0.705$, $0.594 \rightarrow 0.677$). The dependence on k is not monotone: components 9–16 lower test macro-F1 for all three classifiers before later components recover it, and the best MLP value is at $k = 128$ (0.717 validation). Sixteen principal components are therefore not enough for per-pixel classification of these spectra by these classifiers. The top 16 components carry 81.7% of the standardized variance (23: 86.5%; 64: 97.7%).

E.4 STAGE B: JOINT TRAINING ON NATURAL PATCHES

The model is that of Experiment 3 with four logits: linear encoder $942 \to 12$ (or $k \to 12$ in the compression arms, the projection applied to all nine pixels), ReLU patch head of width $M = 32$, class-weighted mean cross-entropy (weights proportional to inverse class frequency, normalized to mean one), full-batch plain gradient descent with one global rate 10^{-3}, a budget of $40{,}000$ steps, paired initialization within seed, three seeds. Arms: *natural* context; *natural, 16 PCs* (inputs projected onto the top 16 training-centre principal directions); *shuffled* (the eight neighbours replaced by random training centres drawn independently of the centre label: the uninformative-context comparator, jointly trained); *shuffled, 16 PCs*; *frozen* (random encoder, head trained on natural patches); *natural, $\kappa = 1/256$* (whole-head rate multiplier); and, under amendment 8.5, *no bottleneck* (identity encoder; the head reads the nine raw standardized spectra) with natural and with shuffled context, the analogue of the "fixed" models of O'Leary et al. (2026). Evaluation on validation patients (test reported separately), each centre held fixed: *iid* natural patches; *context-random* (neighbours replaced by random centres from other cores of the same side, independent labels); *homogeneous* (all nine pixels equal to the centre). The probe is a four-class discriminant on the encoder output of centre spectra, fitted on $4{,}000$ held-out training centres. Retraining follows the protocol of Section 4: a fresh paired-initialization head trained for $20{,}000$ steps on context-random training patches from a frozen saved encoder, scored under context-random evaluation. Because the frozen and 16-component arms plateau above every registered matched-fit threshold (loss 0.61 and 0.53 respectively), the registered fallback applies and the primary comparison is at budget end; matched values at loss 0.5 are given in the tables.

Results, non-denoised copy (Tables 18, 19, 20). *P7, insensitivity of the natural-context model to PCA-16 compression: not met.* At budget end the natural-context model scores 0.732 macro-F1 in distribution with the full spectrum and 0.657 with 16 components on validation patients (0.652 and 0.611 on test); the full-spectrum model keeps fitting (loss 0.19) where the compressed one plateaus (0.42). At matched loss 0.5 the two are indistinguishable (0.643 and 0.636). *P8, reliance on the neighbourhood: met.* Under context-random evaluation the natural-context model falls to 0.234 against 0.680 for the shuffled comparator and 0.705 for the per-pixel MLP. *P9, where the information stopped: head-level.* The retrained head recovers 0.681 from the natural-context encoder against 0.688 from the comparator's, and the two encoders' probes are equal (0.672 and 0.670, from 0.52 at initialization); a random encoder yields only 0.494 after retraining. With natural context the encoder learns the class signal exactly as it does without, and the collapse under context-random evaluation is the head's use of same-class neighbours. *P10, compression asymmetry: not established.* The comparator loses 0.111 from PCA-16 ($0.652 \to 0.541$), but so does the natural-context model, so the registered reading (sensitive comparator, insensitive natural-context model) does not hold. *The secondary rate prediction is not met*: slowing the head to $\kappa = 1/256$ lowers the probe (0.619) and the recovered value (0.644), at budget end and at matched loss 0.5 (0.616 against 0.632). With a context that is a redundant copy of the spectral cue there is no competition for the head's rate to relieve.

Results, PCA-23-denoised copy (Tables 21, 22, 23, 24). The replication registered before its cache was built agrees with the non-denoised copy on every verdict. The per-pixel gate passes more strongly (MLP $0.665 \to 0.804$ validation, $0.634 \to 0.883$ test from 16 to 942 components). The natural-context model is again sensitive to PCA-16 at budget end (0.727 against 0.683 in every seed; 0.839 against 0.640 on test) and indistinguishable at matched loss; it collapses under context-random evaluation (0.274); its encoder's probe equals the comparator's (0.671 and 0.676 from 0.529) and a retrained head recovers 0.706–0.712 against the comparator's 0.720; the slow head does not raise the probe consistently (0.683, two seeds up and one down). One clause fails on this copy: the jointly trained comparator (0.722) stays 0.08 below the per-pixel MLP (0.804), as does the no-bottleneck comparator (0.715), so the gap is the head family and plain gradient descent, not the bottleneck.

P11: the learned bottleneck against none. The comparison O'Leary et al. (2026) actually made is a learned linear bottleneck against no reduction, and it reproduces on both copies: with natural context the learned twelve-dimensional bottleneck scores 0.732 in distribution against 0.744 without a bottleneck (denoised copy 0.727 against 0.744; on its test patients the bottleneck model is 0.037 higher), and the shuffled-context comparators score 0.680 against 0.661 under context-random evaluation (0.722 against 0.715 denoised); the no-bottleneck model fits faster, so the equality is not a fitting artefact. Taken with Stage A and P7, this separates two readings of "a small set of spectral features": the class signal is low-dimensional as a learned linear subspace, as any four-class discriminant is, and not as a small number of principal components; the learned features draw on the whole spectrum, so compressibility is not redundancy. Whether a spatial model leaves that information unused is a separate question, answered here in the negative for natural 3×3 context (the model uses the full spectrum and its encoder is not starved) and in the affirmative for the constructed context of Section 5.

Table 17: Experiment 4 (non-denoised copy), Stage A: four-class macro-F1 of per-pixel classifiers on the centre spectra against the number of principal components k (PCA on standardized training-split centres). Shrinkage Fisher discriminant (LDA), balanced logistic regression (LR) and a 256-unit ReLU MLP with a fixed schedule (mean±sd over three seeds), on validation and test patients.

k	LDA (val)	LR (val)	MLP (val)	LDA (test)	LR (test)	MLP (test)
2	0.477	0.482	0.468±0.023	0.479	0.485	0.465±0.024
4	0.506	0.517	0.528±0.011	0.517	0.525	0.520±0.019
8	0.558	0.571	0.616±0.011	0.596	0.606	0.621±0.007
16	0.560	0.595	0.640±0.012	0.538	0.571	0.594±0.009
23	0.592	0.615	0.654±0.004	0.563	0.587	0.612±0.012
32	0.603	0.618	0.674±0.003	0.576	0.597	0.619±0.003
64	0.635	0.656	0.696±0.005	0.607	0.627	0.643±0.002
128	0.664	0.672	0.717±0.006	0.623	0.643	0.677±0.009
256	0.650	0.668	0.703±0.003	0.629	0.655	0.680±0.011
942	0.651	0.673	0.705±0.002	0.645	0.658	0.677±0.007

Table 18: Experiment 4 (non-denoised copy), Stage B, validation patients: mean±sd over seeds (number of seeds) at the matched threshold L^* and at budget end (40,000 steps or training loss 0.10). Probe = macro-F1 of a discriminant on the encoder output of centre spectra; gain relative to initialization. Macro-F1 of the trained model under iid (natural), context-random (neighbours replaced by pixels from other cores with independent labels) and homogeneous (all nine pixels equal to the centre) evaluation.

arm	at	step	probe	probe gain	iid	ctx-random	homogeneous
natural, no bottleneck	L^*	3842±321	0.645±0.010	0.000±0.000	0.719±0.002	0.293±0.008	0.689±0.002 (3)
natural	L^*	13171±676	0.666±0.006	0.144±0.026	0.723±0.005	0.279±0.008	0.681±0.003 (3)
shuffled, no bottleneck	L^*	12248±656	0.645±0.010	0.000±0.000	0.647±0.001	0.657±0.002	0.650±0.008 (3)
natural, no bottleneck	end	30390±1099	0.645±0.010	0.000±0.000	0.744±0.005	0.249±0.007	0.717±0.002 (3)
natural	end	40000±0	0.672±0.006	0.150±0.025	0.732±0.009	0.234±0.003	0.705±0.003 (3)
natural, $\kappa = 1/256$	end	40000±0	0.628±0.015	0.106±0.024	0.652±0.012	0.296±0.018	0.631±0.007 (3)
natural, 16 PCs	end	40000±0	0.574±0.010	0.031±0.020	0.657±0.004	0.201±0.024	0.620±0.007 (3)
shuffled, no bottleneck	end	40000±0	0.645±0.010	0.000±0.000	0.675±0.005	0.661±0.006	0.674±0.008 (3)
shuffled (comparator)	end	40000±0	0.670±0.010	0.148±0.012	0.652±0.008	0.680±0.006	0.648±0.012 (3)
shuffled, 16 PCs	end	40000±0	0.557±0.011	0.015±0.014	0.541±0.004	0.583±0.011	0.540±0.008 (3)
natural, frozen encoder	end	40000±0	0.521±0.019	0.000±0.000	0.576±0.015	0.264±0.022	0.540±0.016 (3)

F Public hyperspectral replication (Pavia University)

F.1 Purpose and registration

Our tissue spectra are not public. This appendix repeats Experiment 3 (constructed context: the mechanism) and Experiment 4 (natural context: usage) on the ROSIS Pavia University scene, a standard public hyperspectral benchmark (610×340 pixels, 103 bands, nine labelled classes, $42{,}776$ labelled pixels), with the same code, protocols, arms and pre-registered predictions, so that every number can be reproduced from the released scripts. The scene, the spatial split, the class pair and the disposition rules were fixed before the first registered run (pre-registration section 9); the readiness scan and the per-pixel sweep are descriptive stages that precede the registered runs, as on tissue.

Table 19: Experiment 4 (non-denoised copy), Stage B, test patients: mean±sd over seeds (number of seeds) at the matched threshold L^* and at budget end (40,000 steps or training loss 0.10). Probe = macro-F1 of a discriminant on the encoder output of centre spectra; gain relative to initialization. Macro-F1 of the trained model under iid (natural), context-random (neighbours replaced by pixels from other cores with independent labels) and homogeneous (all nine pixels equal to the centre) evaluation.

arm	at	step	probe	probe gain	iid	ctx-random	homogeneous
natural, no bottleneck	L^*	3842±321	0.651±0.011	0.000±0.000	0.648±0.007	0.255±0.008	0.630±0.004 (3)
natural	L^*	13171±676	0.614±0.011	0.097±0.027	0.655±0.002	0.242±0.007	0.627±0.003 (3)
shuffled, no bottleneck	L^*	12248±656	0.651±0.011	0.000±0.000	0.604±0.018	0.615±0.004	0.601±0.014 (3)
natural, no bottleneck	end	30390±1099	0.651±0.011	0.000±0.000	0.667±0.002	0.220±0.013	0.630±0.002 (3)
natural	end	40000±0	0.610±0.009	0.093±0.025	0.652±0.004	0.206±0.004	0.618±0.004 (3)
natural, $\kappa = 1/256$	end	40000±0	0.601±0.009	0.084±0.018	0.604±0.003	0.292±0.024	0.580±0.006 (3)
natural, 16 PCs	end	40000±0	0.548±0.007	0.025±0.006	0.611±0.005	0.190±0.016	0.562±0.004 (3)
shuffled, no bottleneck	end	40000±0	0.651±0.011	0.000±0.000	0.616±0.016	0.616±0.007	0.616±0.012 (3)
shuffled (comparator)	end	40000±0	0.616±0.006	0.099±0.011	0.597±0.006	0.628±0.004	0.594±0.014 (3)
shuffled, 16 PCs	end	40000±0	0.535±0.000	0.013±0.012	0.518±0.009	0.557±0.004	0.517±0.009 (3)
natural, frozen encoder	end	40000±0	0.517±0.016	0.000±0.000	0.568±0.014	0.235±0.004	0.518±0.022 (3)

Table 20: Experiment 4 (non-denoised copy), head retraining: a fresh paired-initialization head trained for 20,000 GD steps on context-random training patches from a frozen saved encoder; macro-F1 under context-random and iid evaluation on validation and test patients (mean±sd over seeds; n).

encoder from	saved	ctx-random (val)	iid (val)	ctx-random (test)	iid (test)
natural	random init	0.494±0.036	0.482±0.020	0.481±0.044	0.472±0.049 (3)
natural	at L^*	0.678±0.006	0.667±0.009	0.624±0.005	0.612±0.006 (3)
natural	at end	0.681±0.010	0.673±0.020	0.617±0.008	0.600±0.004 (3)
natural, $\kappa = 1/256$	at end	0.644±0.008	0.634±0.019	0.604±0.002	0.578±0.018 (3)
natural, 16 PCs	at end	0.594±0.012	0.574±0.015	0.568±0.002	0.544±0.016 (3)
shuffled	at end	0.688±0.007	0.660±0.011	0.631±0.007	0.607±0.012 (3)

Table 21: Experiment 4 (PCA-23-denoised copy), Stage A: four-class macro-F1 of per-pixel classifiers on the centre spectra against the number of principal components k (PCA on standardized training-split centres). Shrinkage Fisher discriminant (LDA), balanced logistic regression (LR) and a 256-unit ReLU MLP with a fixed schedule (mean±sd over three seeds), on validation and test patients.

k	LDA (val)	LR (val)	MLP (val)	LDA (test)	LR (test)	MLP (test)
2	0.506	0.511	0.496±0.005	0.512	0.524	0.507±0.006
4	0.533	0.538	0.537±0.014	0.545	0.540	0.538±0.013
8	0.496	0.552	0.601±0.008	0.524	0.597	0.615±0.005
16	0.574	0.598	0.665±0.005	0.543	0.586	0.634±0.007
23	0.624	0.649	0.726±0.007	0.617	0.658	0.691±0.009
32	0.683	0.710	0.773±0.005	0.667	0.722	0.739±0.001
64	0.684	0.708	0.775±0.006	0.697	0.748	0.791±0.006
128	0.764	0.765	0.800±0.002	0.836	0.852	0.850±0.008
256	0.705	0.773	0.802±0.003	0.857	0.859	0.880±0.005
942	0.695	0.773	0.804±0.002	0.807	0.858	0.883±0.005

Table 22: Experiment 4 (PCA-23-denoised copy), Stage B, validation patients: mean±sd over seeds (number of seeds) at the matched threshold L^* and at budget end (40,000 steps or training loss 0.10). Probe = macro-F1 of a discriminant on the encoder output of centre spectra; gain relative to initialization. Macro-F1 of the trained model under iid (natural), context-random (neighbours replaced by pixels from other cores with independent labels) and homogeneous (all nine pixels equal to the centre) evaluation.

arm	at	step	probe	probe gain	iid	ctx-random	homogeneous
natural, no bottleneck	L^*	1380±140	0.756±0.008	0.000±0.000	0.714±0.006	0.302±0.012	0.705±0.006 (3)
natural	L^*	4141±41	0.676±0.017	0.147±0.006	0.725±0.002	0.303±0.014	0.699±0.005 (3)
shuffled, no bottleneck	L^*	6522±247	0.756±0.008	0.000±0.000	0.679±0.008	0.698±0.003	0.676±0.009 (3)
shuffled (comparator)	L^*	16013±1176	0.679±0.008	0.150±0.009	0.689±0.011	0.720±0.005	0.688±0.019 (3)
natural, no bottleneck	end	10371±474	0.756±0.008	0.000±0.000	0.744±0.001	0.280±0.010	0.730±0.003 (3)
natural	end	20853±756	0.671±0.013	0.142±0.002	0.727±0.004	0.274±0.018	0.712±0.004 (3)
natural, $\kappa = 1/256$	end	40000±0	0.683±0.016	0.155±0.015	0.697±0.010	0.299±0.018	0.685±0.013 (3)
natural, 16 PCs	end	40000±0	0.582±0.002	0.024±0.024	0.683±0.006	0.187±0.014	0.637±0.005 (3)
shuffled, no bottleneck	end	34031±890	0.756±0.008	0.000±0.000	0.717±0.018	0.715±0.004	0.715±0.017 (3)
shuffled (comparator)	end	40000±0	0.676±0.002	0.147±0.009	0.703±0.003	0.722±0.004	0.700±0.004 (3)
shuffled, 16 PCs	end	40000±0	0.569±0.000	0.011±0.021	0.564±0.024	0.588±0.007	0.556±0.023 (3)
natural, frozen encoder	end	40000±0	0.529±0.011	0.000±0.000	0.594±0.026	0.261±0.016	0.556±0.015 (3)

Table 23: Experiment 4 (PCA-23-denoised copy), Stage B, test patients: mean±sd over seeds (number of seeds) at the matched threshold L^* and at budget end (40,000 steps or training loss 0.10). Probe = macro-F1 of a discriminant on the encoder output of centre spectra; gain relative to initialization. Macro-F1 of the trained model under iid (natural), context-random (neighbours replaced by pixels from other cores with independent labels) and homogeneous (all nine pixels equal to the centre) evaluation.

arm	at	step	probe	probe gain	iid	ctx-random	homogeneous
natural, no bottleneck	L^*	1380±140	0.889±0.025	0.000±0.000	0.705±0.008	0.274±0.007	0.691±0.005 (3)
natural	L^*	4141±41	0.673±0.009	0.105±0.026	0.702±0.006	0.271±0.015	0.688±0.011 (3)
shuffled, no bottleneck	L^*	6522±247	0.889±0.025	0.000±0.000	0.689±0.002	0.694±0.005	0.680±0.008 (3)
shuffled (comparator)	L^*	16013±1176	0.693±0.006	0.125±0.024	0.698±0.016	0.720±0.004	0.693±0.022 (3)
natural, no bottleneck	end	10371±474	0.889±0.025	0.000±0.000	0.802±0.005	0.248±0.008	0.777±0.001 (3)
natural	end	20853±756	0.737±0.013	0.169±0.029	0.839±0.005	0.226±0.013	0.793±0.008 (3)
natural, $\kappa = 1/256$	end	40000±0	0.674±0.013	0.105±0.008	0.678±0.006	0.293±0.027	0.669±0.010 (3)
natural, 16 PCs	end	40000±0	0.553±0.002	0.007±0.031	0.640±0.003	0.177±0.006	0.582±0.003 (3)
shuffled, no bottleneck	end	34031±890	0.889±0.025	0.000±0.000	0.755±0.018	0.755±0.005	0.748±0.018 (3)
shuffled (comparator)	end	40000±0	0.726±0.002	0.158±0.020	0.763±0.014	0.782±0.003	0.762±0.013 (3)
shuffled, 16 PCs	end	40000±0	0.546±0.002	0.000±0.031	0.556±0.014	0.580±0.005	0.549±0.004 (3)
natural, frozen encoder	end	40000±0	0.568±0.020	0.000±0.000	0.638±0.034	0.234±0.018	0.577±0.031 (3)

Table 24: Experiment 4 (PCA-23-denoised copy), head retraining: a fresh paired-initialization head trained for 20,000 GD steps on context-random training patches from a frozen saved encoder; macro-F1 under context-random and iid evaluation on validation and test patients (mean±sd over seeds; n).

encoder from	saved	ctx-random (val)	iid (val)	ctx-random (test)	iid (test)
natural	random init	0.503±0.018	0.492±0.019	0.523±0.053	0.508±0.072 (3)
natural	at L^*	0.712±0.004	0.703±0.006	0.705±0.005	0.689±0.017 (3)
natural	at end	0.706±0.003	0.701±0.001	0.791±0.005	0.780±0.019 (3)
natural, $\kappa = 1/256$	at end	0.702±0.010	0.705±0.010	0.698±0.010	0.679±0.012 (3)
natural, 16 PCs	at end	0.595±0.011	0.592±0.015	0.584±0.002	0.573±0.015 (3)
shuffled	at L^*	0.726±0.001	0.703±0.012	0.732±0.008	0.716±0.013 (3)
shuffled	at end	0.720±0.005	0.704±0.011	0.782±0.007	0.761±0.003 (3)

F.2 SPATIAL SPLIT, SAMPLING AND PREPROCESSING

The patient-level split of the tissue experiments has no counterpart in a single scene, so we cut the scene into 20×20-pixel tiles and assign tiles at random (seed 0) to training, validation and test in the proportion $0.6/0.2/0.2$ ($208/70/70$ tiles carrying labels), requiring at least 60 admissible pixels per class in each split; a pixel is admissible only if its whole 3×3 neighbourhood lies inside its tile, so no evaluation neighbourhood overlaps a training one ($34{,}386$ admissible pixels). Per tile and class up to $400/200/200$ pixels are sampled with a fixed seed: $19{,}870$ training, $6{,}172$ validation and $6{,}407$ test pixels. Residual spatial autocorrelation across tile borders is a limitation shared by every single-scene benchmark and is not removed here. Reflectance counts are cast to float; per-feature standardization and PCA are fitted on the training split by the experiment code, exactly as for the tissue spectra; there is no per-pixel normalization, denoising or derivative.

F.3 EXPERIMENT 3 ON PAVIA: CONSTRUCTED CONTEXT

The pair is Meadows ($+1$) against Trees (-1), the two largest vegetation classes and a standard confusion in this scene ($8{,}347$ and $1{,}322$ admissible training pixels). The readiness scan reproduces the tissue structure: with standardized inputs a random 12-dimensional encoder reads the contrast at 0.850 against a discriminant of 0.933; after PCA whitening the random encoder reads 0.667 against 0.955 (Table 25), so the same two regimes (high and low initial probe accessibility) apply, with a smaller asymmetry than on tissue. Calibration matched the constructed contextual discriminant to the centre-only discriminant in each regime ($\tau = 1.89$ standardized; 1.66 whitened at $\gamma = 30$ and at $\gamma = 10$). Sizes are scaled to the pair: up to $2{,}400$ balanced training patches, 400 probe-fit pixels, evaluation on all validation and test pixels of the pair. Arms, widths, rate multipliers, seeds, thresholds, conditions and the retraining protocol are those of Appendix D.

F.4 EXPERIMENT 4 ON PAVIA: NATURAL CONTEXT

All nine classes, macro-F1, with the Stage A gate and the Stage B arms of Appendix E. Stage A gives the opposite of the tissue result (Table 26): the best per-pixel classifier gains only 0.020 macro-F1 on validation and 0.009 on test from the components beyond the sixteenth (MLP $0.891 \to 0.911$ and $0.876 \to 0.885$; discriminant $0.767 \to 0.781$ and $0.742 \to 0.759$), so the registered gate fails and sixteen principal components carry essentially all per-pixel information available to these classifiers on this scene, whose bands are strongly correlated. By the registered rule the compression arms are dropped here and Stage B runs the usage arms and the no-bottleneck arms.

Table 25: Pavia University readiness scan (tile-disjoint halves of the training split): centre-only discriminant (oracle) and the discriminant on a random K-dimensional linear encoder's output (mean [min, max] over three encoders), standardized and PCA-whitened inputs; $|\cos|$ is the cosine between the class-mean contrast and the first principal direction.

pair	preprocessing	oracle	$K=2$	$K=4$	$K=12$	$K=32$	$\lvert\cos\rvert$
Meadows vs Trees	standardized	0.933	0.666 [0.62, 0.71]	0.731 [0.68, 0.78]	0.850 [0.84, 0.88]	0.921 [0.91, 0.92]	0.91
Meadows vs Trees	whitened	0.955	0.576 [0.57, 0.58]	0.580 [0.57, 0.59]	0.667 [0.62, 0.70]	0.815 [0.77, 0.87]	0.91
Gravel vs Bricks	standardized	0.853	0.630 [0.61, 0.65]	0.638 [0.62, 0.66]	0.780 [0.73, 0.82]	0.837 [0.82, 0.85]	0.88
Gravel vs Bricks	whitened	0.833	0.519 [0.50, 0.55]	0.559 [0.51, 0.60]	0.612 [0.59, 0.64]	0.676 [0.66, 0.69]	0.88
Asphalt vs Bricks	standardized	0.952	0.846 [0.74, 0.91]	0.898 [0.89, 0.91]	0.922 [0.92, 0.93]	0.944 [0.94, 0.95]	0.99
Asphalt vs Bricks	whitened	0.945	0.544 [0.50, 0.57]	0.566 [0.55, 0.57]	0.650 [0.59, 0.70]	0.775 [0.75, 0.82]	0.99
Meadows vs BareSoil	standardized	0.910	0.750 [0.74, 0.75]	0.779 [0.77, 0.79]	0.837 [0.82, 0.85]	0.908 [0.90, 0.91]	0.99
Meadows vs BareSoil	whitened	0.925	0.605 [0.57, 0.66]	0.662 [0.64, 0.69]	0.729 [0.71, 0.75]	0.797 [0.78, 0.82]	0.99

G SCOPE LIMITATIONS FROM A PRODUCTION MODEL

This appendix reproduces, from the earlier diagnostic draft of this work, the measurements on a production spectral–spatial segmentation model that are summarized in Section 6. They concern initialization curvature and a controlled retraining study; they do not measure the mechanism of Theorem 1 and are included to document why scalar curvature is not the diagnostic used in this paper and why the earlier apparent freezing advantage is not evidence for or against it. The data are breast-tissue QCL micro-array cores with a prostate QCL replication. Theorem and proposition

Table 26: Experiment 4 (Pavia University), Stage A: four-class macro-F1 of per-pixel classifiers on the centre spectra against the number of principal components k (PCA on standardized training-split centres). Shrinkage Fisher discriminant (LDA), balanced logistic regression (LR) and a 256-unit ReLU MLP with a fixed schedule (mean±sd over three seeds), on validation and test patients.

k	LDA (val)	LR (val)	MLP (val)	LDA (test)	LR (test)	MLP (test)
2	0.549	0.572	0.594±0.006	0.719	0.723	0.731±0.003
4	0.639	0.662	0.722±0.004	0.676	0.733	0.776±0.008
8	0.663	0.741	0.790±0.005	0.688	0.810	0.826±0.006
16	0.767	0.816	0.891±0.004	0.742	0.831	0.876±0.003
23	0.783	0.825	0.902±0.004	0.757	0.838	0.882±0.004
32	0.785	0.828	0.905±0.004	0.770	0.839	0.882±0.005
64	0.784	0.829	0.907±0.005	0.762	0.840	0.882±0.003
103	0.781	0.830	0.911±0.005	0.759	0.841	0.885±0.006

numbers quoted inside this appendix refer to the earlier draft's framework and are retained only where the sentence is a scope statement about that framework.

G.1 Production model details

G.1.1 Pipeline, capacities and data

The data are 336×336 pixel tissue-microarray cores collected by quantum-cascade-laser infrared spectroscopy over the 1000–1800 cm^{-1} fingerprint region, with per-pixel four-class tissue labels. Each pixel carries an $S = 942$ dimensional vector, obtained by stacking baseline-corrected absorbance with its first two derivatives. The compositional model is the Block Vision Transformer of O'Leary et al. (2026): a linear per-pixel $f_{\boldsymbol{\theta}} : \mathbb{R}^{942} \to \mathbb{R}^{K}$, then 16×16 patch tokenization, a 12-layer, 12-head vision transformer of embedding width M, a transposed-convolution decoder and a convolutional segmentation head. Parameter counts in Table 27 are read from the instantiated model.

Table 27: Module capacities of the production pipeline, counted from the instantiated model. K is the spectral bottleneck width. The last row is the single-channel field-standard input convention, for comparison.

Configuration	C_f	C_g	C_g/C_f
$K = 64$, three-channel input	61,108	18,271,540	299
$K = 128$, three-channel input	121,588	21,472,564	177
$K = 64$, single-channel input	20,916	18,271,540	874

Channel convention. The published architecture operates on the second-derivative spectrum alone, the field-standard preprocessing in infrared histopathology. The models analysed here instead stack raw absorbance with its first two derivatives, following a companion tokenization study (anonymous, under review). The choice affects only C_f, and the three-channel convention is the conservative one for the present purpose: under the field standard the capacity asymmetry is roughly three times larger. Within the spatial module the dominant blocks are the transformer (5.34×10^6 parameters) and the transposed convolution (9.44×10^6), both $\Theta(M^2)$ in the embedding width.

The head hypothesis is structurally unmet. The segmentation head is the chain $\mathrm{Conv}_{3\times3}(M{+}K \to 96) \to \mathrm{BN} \to \mathrm{ReLU} \to \mathrm{Conv}_{3\times3}(96 \to 48) \to \mathrm{BN} \to \mathrm{ReLU} \to \mathrm{Conv}_{3\times3}(48 \to 4)$, so the final affine classifier sees $48 \times 3 \times 3 = 432$ input features at every width we build ($M \in \{48, 192, 384\}$, $K \in \{64, 128\}$): the $\Theta(M)$ feature map is separated from the logits by two nonlinearities. The curvature-disparity theorem requires a dense classifier over penultimate features of dimension $\Theta(M)$ with no downstream nonlinearity, and that hypothesis is not satisfied here. Failing a hypothesis removes a guarantee; it does not reverse the conclusion.

G.1.2 GEOMETRY AT INITIALIZATION

Gauss–Newton blocks of the instantiated pipeline were measured on real training batches at random initialization, by Lanczos with full reorthogonalization on matrix-free GGN block operators, validated to 10^{-12} against dense computation, sweeping $M \in \{48, 96, 192, 384\}$ with three seeds and four batches per width. Table 28 collects the results.

Table 28: Production geometry at initialization. Ratios are mean ± sd over three seeds and four batches per width. Slopes are log–log fits over the width sweep. Effective ranks are out of $S = 942$. Directional entries are measured at $M = 192$ over the nine measurements whose batch contains both classes (three seeds × three batches).

Quantity	Breast	Prostate
$\mathcal{D}_{\mathrm{curv}}$ at $M = 48$	0.33 ± 0.12	—
$\mathcal{D}_{\mathrm{curv}}$ at $M = 192$	0.46 ± 0.21	—
$\mathcal{D}_{\mathrm{curv}}$ at $M = 384$	0.70 ± 0.19	1.09 ± 0.52
log–log slope of $\lambda_{\max}(G_{\boldsymbol{\theta\theta}})$ vs M	-0.003	-0.012
log–log slope of $\lambda_{\max}(G_{\boldsymbol{\phi\phi}})$ vs M	0.38 (115 $\rightarrow$ 258)	0.45
Effective rank of Σ_X, raw inputs	1.07	1.02
Effective rank of Σ_X, post-normalization	1.21	—
$\lvert \cos \rvert$ between v_1 and the mean spectrum (post-norm. / raw)	$> 0.995 / > 0.9999$	—
$\boldsymbol{\theta}$-block top eigenvector's norm fraction in the v_1-induced subspace	0.954 ± 0.017	—
Trace fraction in the top 40 of 61,108 spectral directions	0.83 ± 0.12	—
Curvature ratio, v_1 vs full class contrast ($\lvert \cos(d, v_1) \rvert = 0.45$)	3.7× (2.6–4.8)	—
Curvature ratio, v_1 vs v_1-orthogonalized contrast	12–28×	—
Squared-norm fraction of the contrast retained after orthogonalization	76–84%	—
Rank at which $\lambda_{\max}(G_{\boldsymbol{\phi\phi}})$ overtakes the $\boldsymbol{\theta}$ spectrum	2–5	—

Two controls bracket the denominator. Removing the spectral normalization deflates $\lambda_{\max}(G_{\boldsymbol{\theta\theta}})$ by 2.4× (471 $\rightarrow$ 194 at $M = 192$) and pushes the ratio across parity (1.18 at $M = 192$, 1.31 at $M = 384$). Input *centering* changes nothing when the architecture contains internal batch normalization: $\mathcal{D}_{\mathrm{curv}}$ is identical to three decimals with and without centering, because a linear projection followed by normalization absorbs constant input shifts exactly. The near-rank-one input statistics make the cap side of the curvature-disparity bound uninformative at buildable widths independently of the head-hypothesis failure: the data concentrate almost all their second-moment mass in one direction, and that direction is the mean spectrum. The directional measurements establish an anisotropy, not its consequence for learning; whether it starves learning is a hypothesis, not a measurement.

G.1.3 WHAT CARRIES THE WIDTH TREND

The measurements in this subsection are made on the production model at random initialization, at three widths $M \in \{48, 192, 384\}$ and three seeds, on one core (the densest of the first 24 breast fold-0 training cores, with 29,664 labelled pixels of 112,896), with cuDNN TF32 disabled. Four normalization modes are compared: `train` (both head batch normalizations using batch statistics), `eval_bn1` and `eval_bn2` (only the first or only the second head normalization at its initial running statistics) and `eval_head` (both). Artifacts: `results/exp1_8c_interior_witness.csv` and `results/exp1_8d_REPORT.md`.

The incoming features are flat in width. The only downstream layer whose input dimension grows with M is the first head convolution. The top eigenvalue of the patch-unfolded second-moment matrix of the features entering it is unchanged to four significant figures across the three widths, at 356.3, 403.7 and 332.4 for seeds 0, 1 and 2 respectively; the leading direction sits inside the width-independent $K = 64$ skip channels. Table 29 also records a discrepancy that matters for any normalization argument: the same Gram computed over the full batch-normalization group (all positions, ignored pixels included) rather than over labelled pixels is substantially smaller, and the cosine between the two centered top eigenvectors is far from one.

Normalization quantities. Table 30 gives the per-channel batch statistics of the two head normalizations, measured over the group they actually normalize, together with the activation energy

Table 29: Incoming 3×3-patch Gram of the first head convolution, top eigenvalue, from `results/exp1_8d_REPORT.md`. Values are identical across $M \in \{48, 192, 384\}$ and across all four normalization modes, so a single column suffices. *labelled* uses the labelled-pixel mask; *full group* uses the batch-normalization group. The last column is the cosine between the two centered top eigenvectors.

seed	labelled	labelled, centered	full group	cos(labelled, full)
0	356.3	68.19	263.3	0.2793
1	403.7	54.60	306.8	0.5502
2	332.4	67.14	253.0	0.2176

entering and leaving the first one. Under fan-in initialization the pre-normalization variance of the first head normalization falls roughly as $1/(M+K)$, and its inverse-standard-deviation gain grows correspondingly. No channel's variance is within a factor of ten of the stabilizer $\varepsilon = 10^{-5}$: the fraction of channels with variance below 10ε is 0 in every row of both normalizations at every width, seed and mode. (This recorded check does not establish the stronger separation by three orders of magnitude claimed in the earlier draft.)

Table 30: Head normalization quantities at initialization, `train` mode, from `results/exp1_8d_REPORT.md` Table 1. Per-seed values are per-channel medians (variance, gain) or labelled-pixel energies; the *mean* column is the mean of the three seed values, computed by us. $\mathrm{gain}_i = \gamma_i / \sqrt{\mathrm{var}_i + \varepsilon}$. The last row is the product of the mean variance with $M + K$, $K = 64$.

Quantity	seed 0	seed 1	seed 2	mean	M
First head BN, median variance	0.1470	0.1467	0.1587	0.1508	48
	0.0678	0.0589	0.0656	0.0641	192
	0.0346	0.0310	0.0398	0.0351	384
First head BN, median gain	2.609	2.611	2.511	2.577	48
	3.842	4.120	3.905	3.956	192
	5.375	5.682	5.010	5.356	384
Second head BN, median gain	2.892	3.122	3.039	3.018	48
	3.110	3.177	3.195	3.161	192
	2.966	3.060	3.121	3.049	384
Pre-BN energy, labelled pixels	25.64	26.28	23.91	25.28	48
	11.07	10.67	11.19	10.98	192
	5.765	5.543	6.015	5.774	384
Post-BN energy, labelled pixels	132.1	130.8	133.2	132.0	48
	130.4	131.5	132.5	131.5	192
	130.7	128.5	130.8	130.0	384
Mean variance $\times\,(M{+}K)$	—	—	—	16.9 / 16.4 / 15.7	48 / 192 / 384

Witness Jacobian-vector products. A unit perturbation of the first head convolution is propagated by forward-mode differentiation through the actual head in the actual normalization mode, so that the train-mode derivative includes the dependence on the batch statistics. The final scalar $e_{\text{final}} = N^{-1} \sum_n d_n^\top (\mathrm{diag}(p_n) - p_n p_n^\top) d_n$ is a lower-bound witness for $\lambda_{\max}(G_{\boldsymbol{\phi\phi}})$ and is verified against the same matvec the eigensolver uses; all 216 rows pass at relative discrepancy below 10^{-3}, with a maximum of 1.269×10^{-5} over the 198 rows with nonzero curvature. Table 31 gives the stage energies for two directions in two modes.

A spatially constant bias perturbation $\Delta b = e_c$ is annihilated exactly by the immediately following train-mode normalization: in `train` and `eval_bn2` modes, at every width and seed, all stage energies after the centering step are exactly zero and the quadratic form agrees with the matvec to at most 1.14×10^{-17} in absolute value. In `eval_head` the same perturbation gives e_{final} between 7.8×10^{-4} and 2.0×10^{-3}, and in `eval_bn1` between 0.0154 and 0.0692.

Slopes and attribution. Table 32 gives the log–log slopes against M per seed and pooled, together with the mode-to-mode ratios of levels.

Table 31: Witness JVP stage energies, mean over three seeds, from `results/exp1_8d_REPORT.md` Table 2a. `ggn_top` is the full top GGN eigenvector of the convolution block; `s9cen` is the centered leading direction of the incoming patch Gram. e_{pre} is the perturbation energy before the first head normalization, e_{logit} is measured on labelled pixels.

Mode	Direction	M	e_{pre}	e_{bn1}	e_{bn2}	e_{logit}	e_{final}
`train`	`ggn_top`	48	260.6	2875	4453	261.7	60.73
`train`	`ggn_top`	192	259.4	5905	7553	505.2	118.5
`train`	`ggn_top`	384	256.8	9168	13810	1018	227.6
`train`	`s9cen`	48	76.53	1538	1224	29.23	5.761
`train`	`s9cen`	192	76.53	2305	1914	50.83	9.862
`train`	`s9cen`	384	76.53	3997	3906	124.3	21.95
`eval_head`	`ggn_top`	48	261.9	261.9	42.42	9.607	2.366
`eval_head`	`ggn_top`	192	262.8	262.8	42.18	10.35	2.487
`eval_head`	`ggn_top`	384	262.5	262.5	46.53	10.45	2.372
`eval_head`	`s9cen`	48	76.53	76.53	9.103	0.4667	0.1049
`eval_head`	`s9cen`	192	76.53	76.53	8.460	0.4613	0.1013
`eval_head`	`s9cen`	384	76.53	76.53	9.762	0.4450	0.1011

Table 32: Log–log slopes against M over the three widths, per seed and pooled, and mode-to-mode ratios of the levels (each mode's value divided by its train-mode value at the same width and seed, then averaged over seeds and over the three widths), from `results/exp1_8d_REPORT.md`. λ_{WW} is the top curvature of the first head convolution's weight block; λ_{Φ} is the whole spatial block. Three-width slopes are descriptive, not asymptotic exponents.

	slope of λ_{WW}				λ_{Φ}	level ratio vs `train`	
Mode	seed 0	seed 1	seed 2	pooled	pooled slope	λ_{WW}	λ_{Φ}
`train`	0.509	0.785	0.350	+0.613	+0.398	1	1
`eval_bn1`	0.429	0.634	0.542	+0.563	+0.281	0.903	1.03
`eval_bn2`	0.544	0.632	0.508	+0.571	+0.253	0.156	0.227
`eval_head`	−0.112	0.181	−0.084	+0.006	−0.331	0.026	0.037

Three readings, and one non-reading. The positive width trend in the spatial block's curvature at initialization is sensitive to head batch normalization: holding both head normalizations at their initial running statistics removes it ($+0.613 \to +0.006$ for the convolution block, $+0.398 \to -0.331$ for the whole spatial block) while leaving the incoming features unchanged. Either normalization alone preserves the trend ($+0.563$ and $+0.571$), which is evidence against attributing it uniquely to the first one, even though most of the drop in the *level* attaches to the second. The mechanism this supports is a finite-range normalization gain acting on width-diluted activations, over the range of widths measured, and not the width-linear feature-Gram mechanism of the shallow theory. The non-reading: the `eval_head` intervention changes the normalization derivative, the forward logits, the downstream gates and the cross-entropy metric at the same time, so a difference between modes establishes sensitivity to the intervention, not a unique cause; and the observation that the variance never approaches the stabilizer is a heuristic about the regime, not a guarantee.

Parameterization control. Table 33 scales the weights and bias of the first head convolution by c and, in the exact invariance, the following stabilizer by c^2. The logits are unchanged and the block's top curvature scales by exactly c^{-2}: the raw Euclidean curvature of a layer feeding a normalization is a parameterization quantity, not an architectural one (van Laarhoven, 2017).

Table 33: Function-preserving scale control at seed 0, from `results/exp1_8d_REPORT.md` Table 3. *ratio* is $c^2\lambda_{WW}(c)/\lambda_{WW}(1)$, which is 1 under exact invariance. *logit dev.* is $\max|\text{logits}(c) - \text{logits}(1)|$, reported both with the stabilizer co-scaled by c^2 and with it held fixed (the implementation's actual behaviour); λ_{WW}, *ratio* and $\|W\|_F^2$ are taken from the fixed-stabilizer variant, and the co-scaled variant differs from it by at most one unit in the fourth significant figure.

M	c	λ_{WW}	ratio	logit dev. (co-scaled)	logit dev. (fixed ε)	$\|W\|_F^2$
48	1/4	719.4	1.000	0	$9.40 \cdot 10^{-4}$	2.003
48	1/2	179.8	1.000	0	$1.89 \cdot 10^{-4}$	8.014
48	1	44.96	1.000	0	0	32.06
48	2	11.24	1.000	0	$4.63 \cdot 10^{-5}$	128.2
48	4	2.810	1.000	0	$5.84 \cdot 10^{-5}$	512.9
192	1/4	1935	0.9998	0	$3.00 \cdot 10^{-3}$	2.002
192	4	7.560	1.000	0	$1.89 \cdot 10^{-4}$	512.6
384	1/4	1927	0.9991	0	$4.69 \cdot 10^{-3}$	1.999
384	4	7.535	1.000	0	$2.95 \cdot 10^{-4}$	511.8

Eigensolver accuracy. Every row of the width sweep terminates on the combined criterion (relative change of the Rayleigh quotient below tolerance *and* residual below 10^{-3}), never on the iteration or wall-clock budget. Over all 36 (mode, width, seed) rows the relative residual $\|Gv - \lambda v\|/\lambda$ is at most 3.81×10^{-4} for the convolution weight block, at most 3.81×10^{-4} for the weight–bias block and at most 9.97×10^{-4} for the whole spatial block, and the Rayleigh quotient of the returned iterate agrees with the reported eigenvalue to the printed precision in every row.

G.1.4 Controlled retraining: original protocol

The pipeline was retrained on breast fold 0 at reduced depth (six transformer layers) with full instrumentation. Two protocols were run and both are reported: the *original* protocol (39 runs), which is the pre-registered one and whose joint-versus-frozen contrast an audit of the training code later showed to be confounded, and the *matched* protocol (21 runs), which removes the confounds, saves the selected checkpoint and re-runs the pre-registered comparisons. Sixty logged runs in total. Per-step residuals, block gradient norms, per-epoch parameter displacements and input-Jacobian norms are recorded in both; the matched protocol additionally logs the per-step clipping coefficient and the per-block applied update norms.

The original protocol comprises arms {joint-linear, frozen-random, frozen-PCA, joint-MLP} $\times$ widths $M \in \{48, 192\}$ $\times$ three seeds under AdamW for 60 epochs, plus a {joint-linear, frozen-random} $\times$ three-seed pair under momentum SGD at both widths and a frozen-PCA SGD triple at $M = 48$ as a positive control. The spectral parameters sit in a zero-weight-decay group in every arm, and frozen arms log counterfactual spectral gradients. Table 34 gives the AdamW arms.

Table 34: Controlled study, original protocol, AdamW arms: validation macro-F1 (mean ± sd over three seeds) under two checkpoint policies, and relative spectral displacement at end of training. *Final-5* is the pre-registered metric. *Best-val* is the maximum over the 60 per-epoch evaluations on the same 28 validation cores; no checkpoint was saved at that maximum and this study has no test set, so that column is exploratory and selection-optimistic. The two policies order the arms oppositely. These arms also differ in clip scope and in normalization-affine treatment, so this contrast is confounded.

	Final-5 (pre-registered)		Best-val (exploratory)		
Arm	M=48	M=192	M=48	M=192	$\|\Delta\boldsymbol{\theta}\| / \|\boldsymbol{\theta}_0\|$
Joint (linear)	0.538 ± 0.082	0.646 ± 0.050	0.873 ± 0.030	0.858 ± 0.056	0.54–0.64
Frozen random	0.648 ± 0.063	0.731 ± 0.023	0.840 ± 0.025	0.817 ± 0.017	0
Frozen PCA	0.731 ± 0.003	0.726 ± 0.012	0.769 ± 0.021	0.862 ± 0.068	0
Joint (MLP)	0.566 ± 0.070	0.575 ± 0.184	0.900 ± 0.000	0.897 ± 0.001	0.85–0.96

On the pre-registered final-window metric the joint-linear minus frozen-random gaps are -0.110 at $M = 48$ (90% paired-t interval $[-0.333, +0.114]$) and -0.085 at $M = 192$ ($[-0.185, +0.015]$), neither resolvable at $n = 3$; against frozen-PCA they are -0.193 ($[-0.334, -0.052]$) and -0.081 ($[-0.175, +0.014]$). Taking each run's maximum over its 60 evaluations reverses the ordering: joint minus frozen-random becomes $+0.033$ and $+0.041$, joint minus frozen-PCA $+0.104$ and -0.004. The switch is an arithmetic identity — the change in every gap equals the difference in peak-to-final degradation between the two arms — and averaged over seeds that degradation is 0.335 ($M = 48$) and 0.212 ($M = 192$) for joint-linear against 0.192 and 0.086 for frozen-random.

Three asymmetries. First, gradient-norm clipping at 1.0 was computed over the optimizer's parameters, that is over $\boldsymbol{\theta} + \boldsymbol{\phi}$ in the joint arms but over $\boldsymbol{\phi}$ alone in the frozen arms, so at identical gradients the frozen arm takes the larger $\boldsymbol{\phi}$ step: the median extra clip factor paid by the joint arm is 1.10–1.12 under SGD at $M = 48$ and 3.0–3.6 under AdamW at $M = 192$. Second, the normalization following the spectral projection contributes 128 affine parameters that are trained in joint-linear, frozen in frozen-random and structurally absent in frozen-PCA. Third, no checkpoint was saved during training, so the best-validation column is a post-hoc curve maximum that no saved model realizes, taken over 60 evaluations of the same 28 validation cores that define the comparison.

G.1.5 Momentum-SGD controls and normalization recalibration

The two optimizer families constitute an optimizer-family control; the SGD arm is not an instantiation of the two-timescale theorem's gradient flow. It uses momentum 0.9; weight decay 0.01 on $\boldsymbol{\phi}$ and 0 on $\boldsymbol{\theta}$; mini-batches of four cores with reshuffling; and global gradient-norm clipping at 1.0, which binds on 73–89% of steps. Logged gradient norms are recorded before clipping, so the plotted theory quantities are not the quantities the trajectory applied. Under momentum SGD the paired joint minus frozen-random gap is $+0.002$ at $M = 48$ and $+0.011$ at $M = 192$ on best-validation, with 90% paired-t intervals $[-0.008, +0.011]$ and $[-0.007, +0.030]$; on the final-window metric the same pairs give $+0.028$ and -0.057, with intervals $[-0.173, +0.228]$ and $[-0.124, +0.010]$ — gaps that differ in sign between the widths, with intervals an order of magnitude wider than the effect they bracket. Both arms lose 0.12–0.22 macro-F1 between their peak and their final window. Descriptively, the spectral share of the summed squared gradient norms drops from about 0.94 (AdamW joint) to about 0.26 (SGD joint); momentum, clipping, weight decay and adaptivity all differ between the two families, so no single factor is isolated.

Recalibration of normalization statistics. On disposable copies of the final checkpoints, with all weights fixed, the normalization running statistics were re-estimated on training data only (29 batches) and the model re-evaluated. For this intervention and these checkpoints, the recovered fraction of the joint arms' peak-to-final drop is 0.17 at $M = 48$ and 0.07 at $M = 192$: recalibration recovers a nonzero part of the degradation but not most of it. What it does undo is the instability of the endpoint number: it repairs the two collapsed joint checkpoints ($+0.24$ and $+0.15$ macro-F1) and shrinks the seed standard deviation by roughly $3.7\times$, while leaving the frozen arms essentially unchanged ($|\Delta| \leq 0.02$, and ≤ 0.006 for frozen-PCA, whose spectral stage carries no normalization).

G.1.6 MATCHED PROTOCOL

The matched protocol equalizes the three asymmetries and re-runs the pre-registered comparisons at $M = 48$: the reduction-stage normalization affine parameters are placed in $\boldsymbol{\phi}$ and trained wherever that normalization exists; the clip norm at 1.0 is computed over $\boldsymbol{\phi}$ only under the same clipping rule and scope in every arm; and the best-validation checkpoint is saved. AdamW at learning rate 10^{-4}, 60 epochs, three seeds, flip and 90°-rotation augmentation on in all arms. The logged per-step clip coefficient averages 0.33–0.57 across arms — the coefficients and gradients differ by arm, which is an outcome of the matched rule — and the logged applied update norms confirm that the spectral share of the applied update tracks the learning-rate multiplier (0.02, 0.17 and 1.36 times the $\boldsymbol{\phi}$ update norm for $\times 0.1$, $\times 1$ and $\times 10$). Table 35 gives all seven arms.

Table 35: Controlled study, matched protocol ($M = 48$, AdamW, three seeds, mean $\pm$ sd). Identical normalization-affine treatment and the same $\boldsymbol{\phi}$-only clipping rule and scope in every arm; best-validation checkpoints saved. *Final-5* is the pre-registered metric; *best-val* is the maximum over 60 per-epoch evaluations on the 28 fold-0 validation cores and is exploratory and selection-optimistic, since this study has no test set. The drift column is the relative spectral-parameter displacement at end of training; its standard deviations were computed by us from `results/e3c_analysis_runs.csv`. Rows 1–2 and rows 3–4 are the two matched pairs; see the text.

Arm	Final-5 (pre-registered)	Best-val (exploratory)	$\lVert\Delta\boldsymbol{\theta}\rVert \,/\, \lVert\boldsymbol{\theta}_0\rVert$
Frozen random	0.684 ± 0.037	0.814 ± 0.040	0
Joint (linear)	0.694 ± 0.056	0.837 ± 0.029	0.566 ± 0.026
Frozen pretrained	0.690 ± 0.033	0.861 ± 0.022	0
Fine-tuned pretrained	0.710 ± 0.052	0.857 ± 0.009	0.020 ± 0.005
Joint, spectral lr $\times 0.1$	0.626 ± 0.055	0.841 ± 0.007	0.083 ± 0.006
Joint, spectral lr $\times 10$	0.561 ± 0.114	0.886 ± 0.037	4.910 ± 0.402
Joint, cosine schedule	0.660 ± 0.020	0.825 ± 0.033	0.359 ± 0.013

Two matched pairs, not a single-factor design. The four core arms do not all differ in one factor. Joint-linear and frozen-random use a linear reduction stage with normalization; frozen-pretrained and fine-tuned-pretrained use a pretrained multilayer spectral encoder without reduction-stage normalization. The clean freezing comparisons are therefore the two matched pairs, joint-linear against frozen-random and fine-tuned against frozen-pretrained. Comparing across the pairs — frozen-pretrained against frozen-random, for instance — changes architecture, initialization and training history, and validation selection, not only informativeness.

Verdicts. All comparisons in Table 36 are paired by seed with 90% t-intervals ($t = 2.920$, $n = 3$), conditional on one validation split, and are descriptive seed summaries rather than a population or multi-factor causal analysis.

Table 36: Matched protocol: paired differences with 90% paired-t intervals ($n = 3$). The last two rows compare the two protocols and are reported as descriptive seed summaries; saving a checkpoint changes what is available to evaluate, not the final-window trajectory.

Contrast	Final-5	Best-val
Fine-tuned − frozen-pretrained	$+0.021\ [-0.022, +0.063]$	$-0.004\ [-0.048, +0.039]$
Joint-linear − frozen-random	$+0.011\ [-0.142, +0.163]$	$+0.023\ [-0.066, +0.112]$
Frozen-pretrained − frozen-random	—	$+0.047\ [-0.042, +0.136]$
Spectral lr $\times 10$ − $\times 1$	$-0.133\ [-0.419, +0.152]$	$+0.048\ [-0.016, +0.112]$
Spectral lr $\times 0.1$ − $\times 1$	$-0.068\ [-0.173, +0.037]$	$+0.004\ [-0.039, +0.046]$
Joint/frozen gap: matched − original	$+0.120\ [-0.230, +0.470]$	$-0.010\ [-0.105, +0.085]$

The pre-registered prediction that fine-tuning a pretrained spectral encoder falls below leaving it frozen is not supported: neither difference is resolvable at $n = 3$, and the fine-tuned arm moved $\boldsymbol{\theta}$ by only 2% of its initial norm. The logged applied update norms verify nonzero applied encoder updates under AdamW in that arm; this is a statement about the applied updates, not an attribution of the loss decrease. The pre-registered harmful-drift ordering $\times 0.1 \geq \times 1 \geq \times 10$ is not observed: on best-validation the $\times 10$ arm is above $\times 1$ in $3/3$ seeds while its final-window endpoint is the lowest and most variable of any arm at 0.561. Three multiplier means do not establish a systematic relation between displacement and peak score, and the intervals are wide. The cosine schedule improved neither mean endpoint (0.825 best-val and 0.660 final-5 against 0.837 and 0.694 for the constant learning rate); one schedule that did not help does not rule out an excessive late learning rate.

Disclosures. Width 48 only; $n = 3$; a single validation fold of 28 fold-0 cores; no test set. The best-validation column is exploratory and the final-window column is the pre-specified metric. The pretrained encoders were selected by pixel validation macro-F1 on the *same* fold-0 validation cores the runner later scores, which is a selection advantage for both pretrained arms, and were pretrained with weight decay 0.01 on $\boldsymbol{\theta}$. Pixel pretraining used the unaugmented centre-crop protocol while subsequent training used random crop plus augmentation; this affects comparisons across the two pairs, not the frozen-versus-fine-tuned pair that starts from the same checkpoint. Augmentation is on in all arms. Finally, the summary artifact `results/e3c_analysis_runs.csv` does not expose per-run configuration hashes or data-order pairing, so before the pairs are described as completely matched a compact provenance manifest should accompany it, recording architecture, initial checkpoint hash, parameter-group membership, clipping and weight-decay settings, normalization mode and state, augmentation and data-order seed, scheduler and selection rule; identical seed labels alone do not establish bit-identical mini-batch order across different model constructions.

G.1.7 What is not certified on this pipeline

The spatial module's input-Jacobian operator norm grows over training, from 3.2 to 157 on average for joint-linear (up to 414) and from 1.7 to 1.2×10^4 for joint-MLP, which rejects an initialization-scale stability approximation for the containment constant of the displacement theorem, though not the existence of a finite-horizon constant over the training window. The fitted per-step residual-envelope rate for joint-linear grows from $\hat{\mu} = 6.0 \times 10^{-4}$ ($M = 48$) to 1.9×10^{-3} ($M = 192$), a factor 3.2 for a $4\times$ width increase; the ratio is invariant to the step-clock-to-flow-time conversion, the absolute values are not, and two widths do not establish a trend.

H Verification record

The statements of this paper rest on the following checks, each a script in the accompanying repository with fixed seeds and a recorded outcome; the record is scoped to these statements and does not certify anything else. (i) Theorem 1: the closed forms, bounds and the normalized-readout control were verified against a full $(2 + M)$-parameter integration of the flow over 525 non-trivial initialization–width–margin cases and a $4{,}000$-draw Gaussian initialization sweep (`code/experiments/toy_serial_ce.py`, `toy_serial_ce_isotropic.py`; Appendix B); the algebraic identities of the proofs were checked symbolically and numerically (`code/audits/astra_math_02_checks.py`). (ii) Experiment 2: the matrix-product implementation of the convolutional head agrees with `nn.Conv2d` with circular padding to 10^{-15} in double precision, in outputs and parameter gradients, at widths 2, 37 and 512 (`exp2_intervention.py -selftest`); the calibration is bitwise reproducible before and after the paired-condition change to the generator; the pre-registered constants, thresholds, predictions and every amendment are dated in `paper/REFOCUS_PLAN_2026-09-10.md`. (iii) Experiment 3: preprocessing, principal directions and the readiness scan are computed on training patients only and evaluated on a core-disjoint half; no test-patient quantity was used for any choice. (iv) The production-model measurements of Appendix G carry their own verification record from the earlier draft (`code/audits/README.md`); only the statements quoted in Section 6 are relied upon here.